



Particle detection Timepix pixel detectors (part I)

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CERN Medipix Collaboration



- ◆ Development of pixel detectors
- ◆ Application of pixel detectors for scientific research
- ◆ Technology transfer initiative
- ◆ Application of pixel detectors in the field of education



Medical Imaging

The Medipix chips provide many advantages for use in the field of medical imaging.



Space Dosimetry

Find out what the Timepix chips are doing at the ISS and in NASA's Orion rocket.



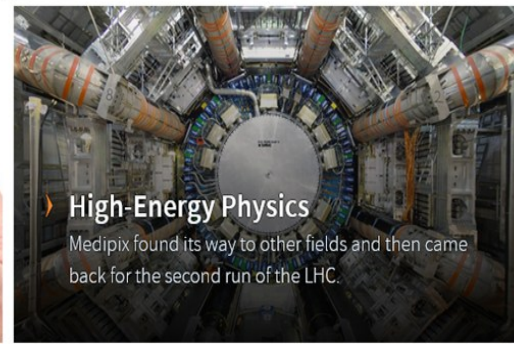
Material Analysis

The Medipix chips are being used for commercial X-ray of materials analysis.



Education

Timepix kits makes up a compelling demonstration tool for educators teaching about radiation.



High-Energy Physics

Medipix found its way to other fields and then came back for the second run of the LHC.

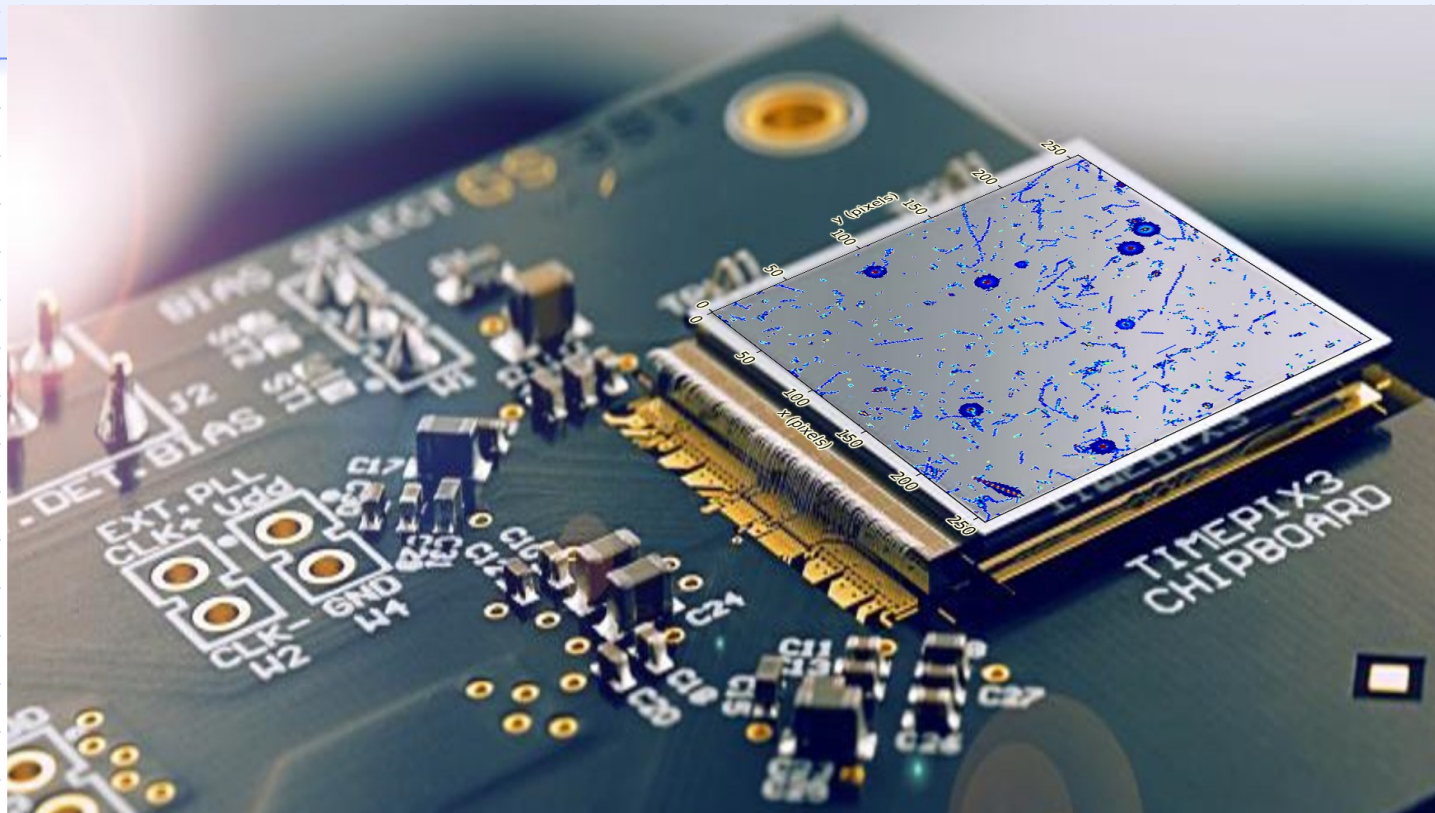


Making Invisible Visible

*....with single energy quantum counting pixel detectors
of Timepix family*



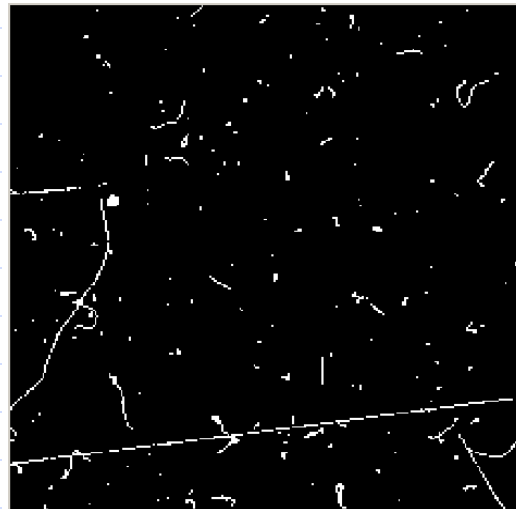
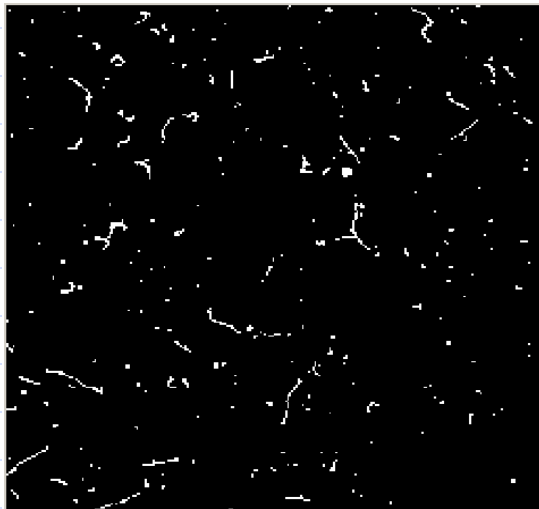
Radiation imaging detector





Pixel detector as "Active emulsion"

- ◆ Some particles can be identified from shape of their tracks as in formerly used nuclear emulsions
- ◆ Example: Cosmic rays and natural background



Are particles photogenic?

Yes of course, they are when observing them with Timepix detectors



Hybrid pixel detector of the Medipix/Timepix family



1998 Medipix1: 64 x 64 pixels, pixel pitch 170 μm

2005 Medipix2 (Mpix2MXR20): First hybrid pixel detector at a pitch <150 μm



Photon counting chips

Optimized for energy sensitive x-ray imaging applications

Each pixel measures the number of interacting photons in few energy bins

2011 Medipix3 RX

2024 Medipix4



Pixel processing chips

Measurement and characterization of single quanta of ionizing radiation

Each pixel can measure energy and/or time of arrival and number of interactions

2006 Timepix

2013 Timepix3

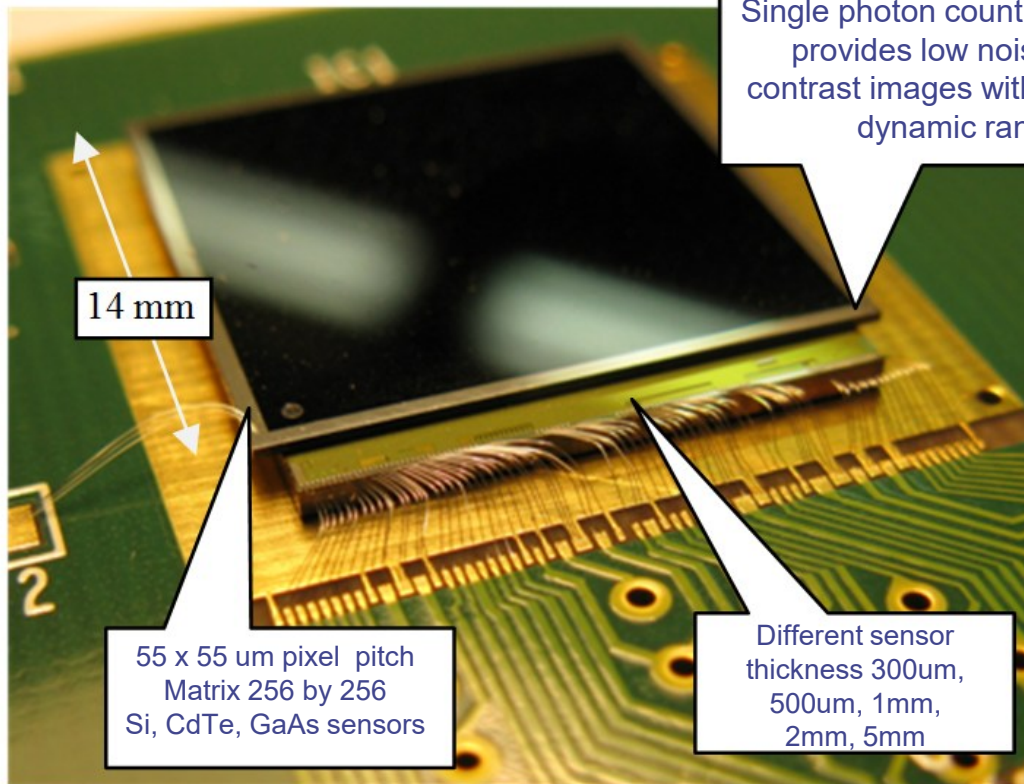
2020 Timepix2

2022 Timepix4





Timepix - Hybrid Pixel Detector with Single Particle Detecting Capability



Single photon counting readout provides low noise, high contrast images with very high dynamic range

14 mm

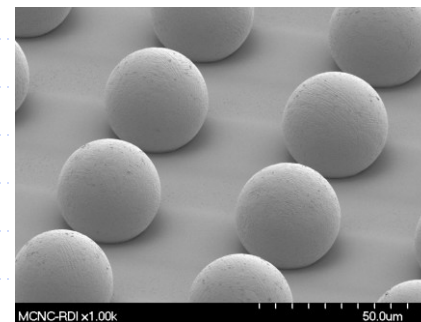
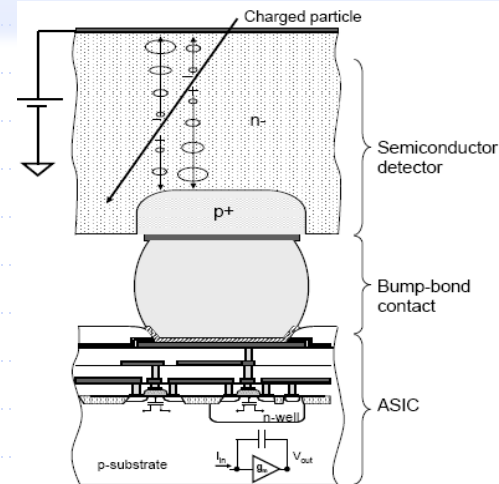
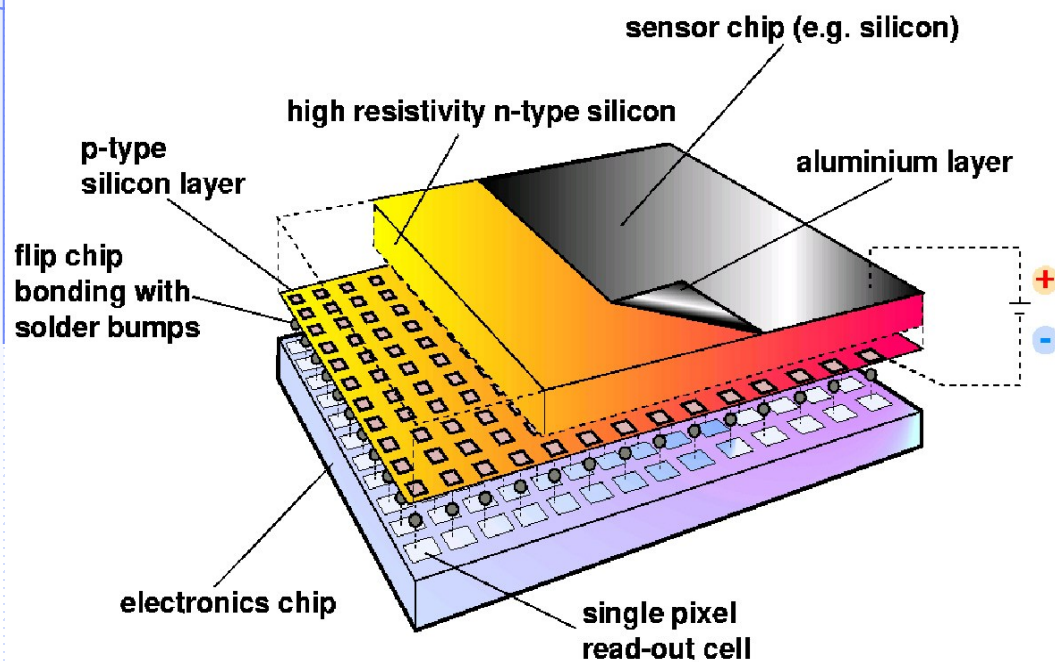
55 x 55 μm pixel pitch
Matrix 256 by 256
Si, CdTe, GaAs sensors

Different sensor thickness 300 μm ,
500 μm , 1mm,
2mm, 5mm

Timepix



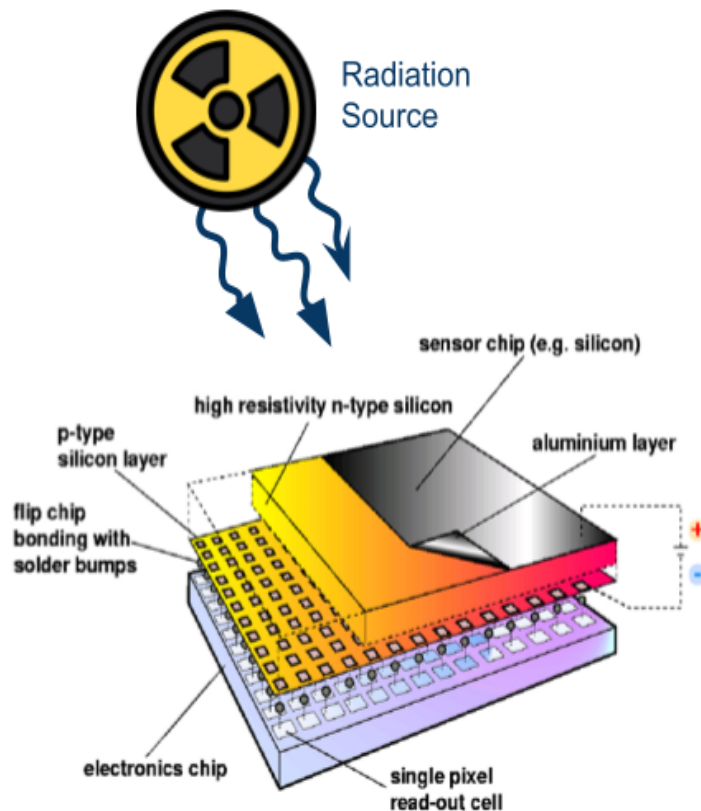
Timepix - Hybrid Pixel Detector with Single Particle Detection Capability



Sensor part and readout electronics part are optimized separately
(sensor materials – Si, CdTe, GaAs)



Particle Camera Working Principle



(1) Dot

Photons and electrons (10keV)

(2) Small blob

Photons and electrons

(3) Curly track

Electrons (MeV range)

(4) Heavy blob

Heavy ionizing particles with low range (alpha particles,...)

(5) Heavy track

Heavy ionizing particles (protons,...)

(6) Straight track

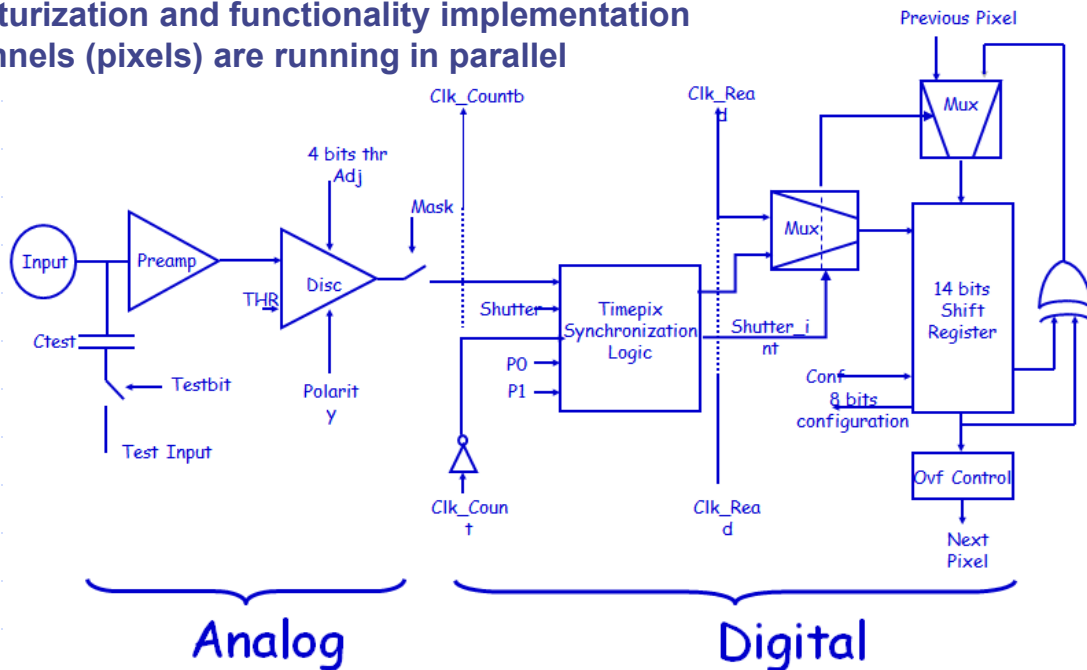
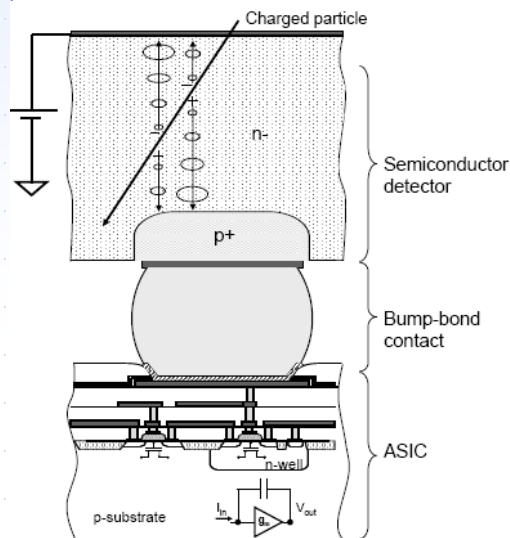
Energetic light charged particles (MIP, Muons,...)



Timepix read-out chip

Single pixel schematics

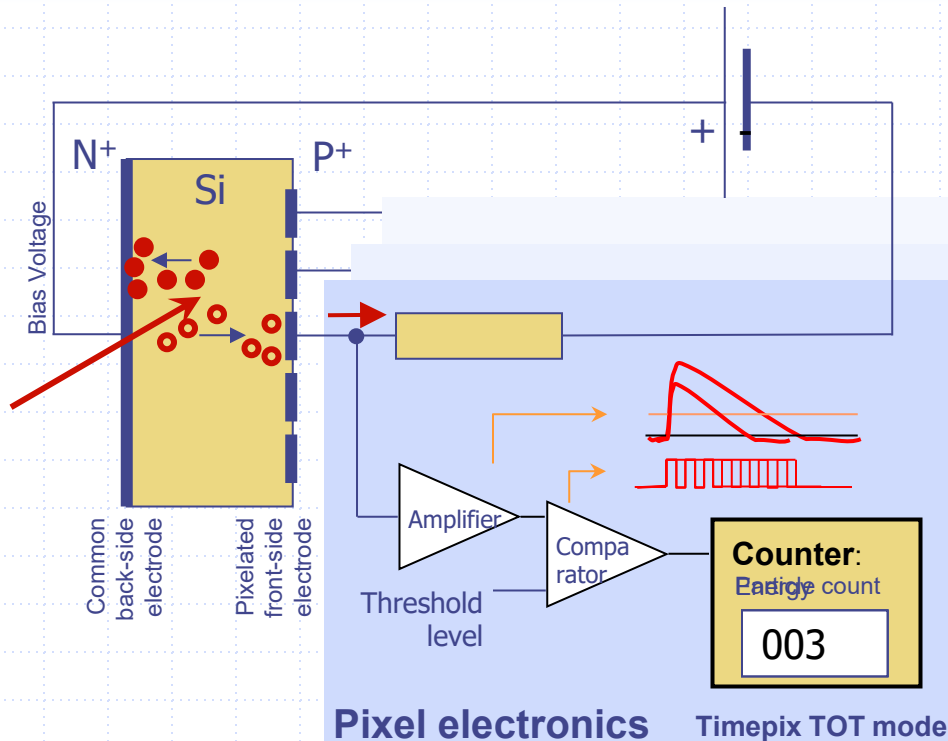
- Independent signal processing is performed in each individual pixel analog (charge sense amplifier, signal shaping) and digital (DA conversion – Wilkinson type converter)
- Very high level of miniaturization and functionality implementation
- All 65 536 detector channels (pixels) are running in parallel





Medipix – single quantum counting detector

Timepix - spectroscopic pixel detector with ToT and ToA modes of operation



Threshold level above electronic noise
⇒ **No false counting.**

Digital integration (counting)
⇒ **No dark current.**



Unlimited dynamic range
and exposure time. Counts obey poissonian distribution.

65k spectroscopic chains:

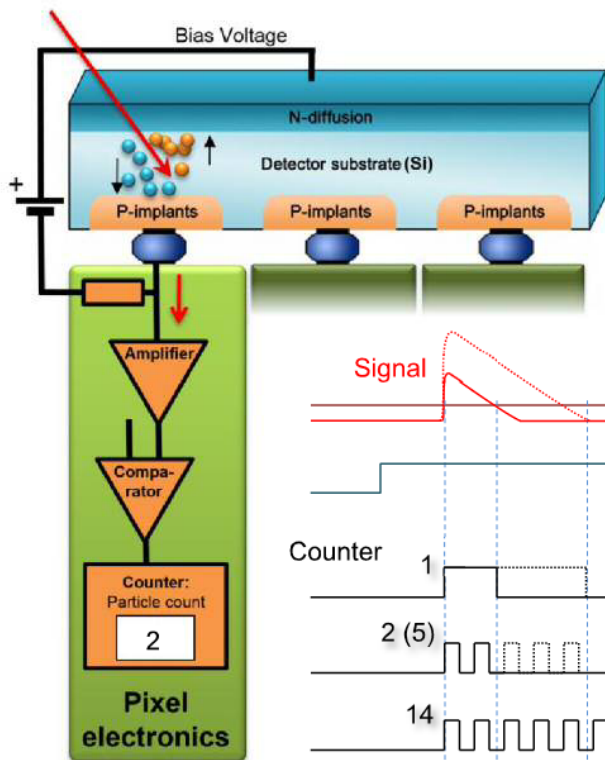
- SCA in case of Medipix
- MCA in case of Timepix

Energy calibration

(defined energy deposited into a volume of pixel, e.g. $55 \times 55 \times 300 \mu\text{m}^3$)



Pixel Operation Modes



◆ Medipix mode

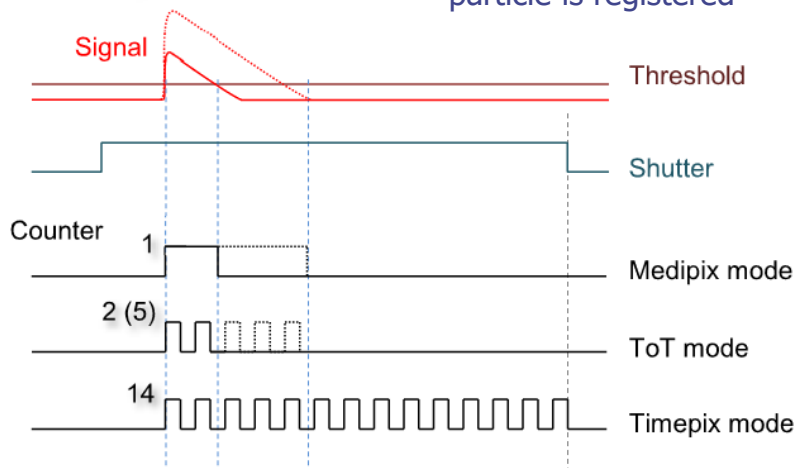
Counter is incremented if incoming particle is registered (value is equivalent to number of detected events)

◆ Time Over Threshold mode

Counter is incremented as long as a signal is over threshold (value is equivalent to energy)

◆ Time of Arrival mode

Counter is continuously incremented from the time when the first particle is registered

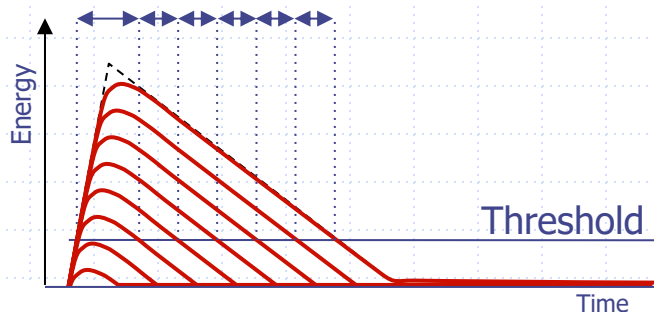
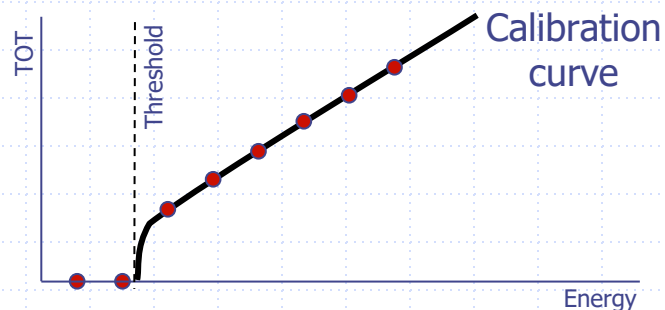




per pixel energy measurement: Time Over Threshold mode

Counter in each pixel can be used as

- Wilkinson type **ADC** to measure energy of each particle detected

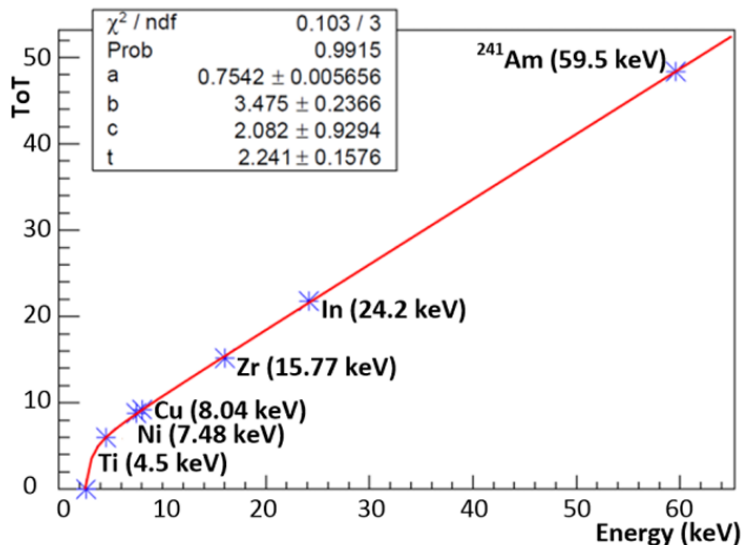


- If the pulse shape is triangular then Time over Threshold is proportional to collected charge i.e. to energy.
- Due to limited bandwidth the pulse can be NEVER perfectly triangular.
- Non-linear TOT to energy dependence
- Threshold defines minimal detectable energy in pixel



Time-over-Threshold Calibration

ToT response on energy deposited in a single pixel



- Each pixel exhibits different response when exposed to same calibration energies
- Minimal detectable energy (threshold) non-zero (typically 2 to 4 keV)

- Sensor is irradiated with photons of known energy
- Single-pixel events are selected and peak position in the energy spectrum is determined

$$f(x) = ax + b - \frac{c}{x-t}$$

Parameters

- ◆ **a** and **b** describing the linear part towards higher energy
- ◆ **c** curvature
- ◆ **t** is related to the threshold



Live Demonstration

Timepix detector
exposed to alpha
radiation



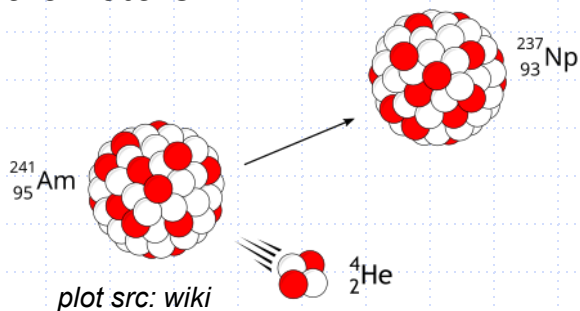


Alpha Particle Detection

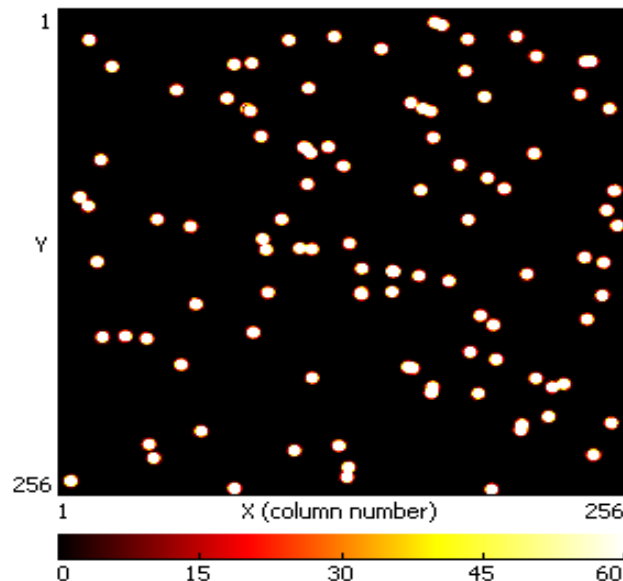
- ◆ Alpha particles represent heavy energetic ions (energy range in MeVs)
- ◆ Alphas are displayed in the pixel matrix as tracks of „heavy blob“ category

*Note that pixel dimensions
are 55 x 55 μm
..and range of alphas in
silicon is just tens of
micrometers*

*How is it possible that
alphas are displayed as
heavy blobs covering
area several tens of
pixels???*



*MinipixEDU (Timepix)
exposed to Am241*

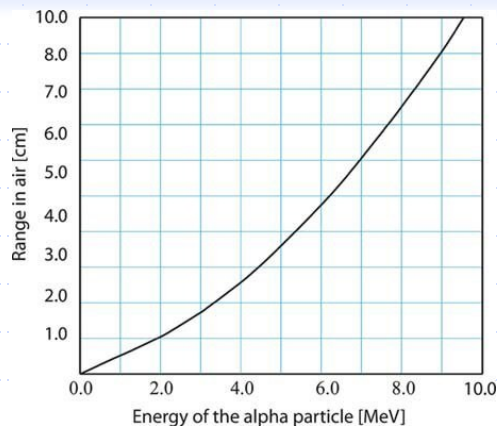




Range of Alpha Particles



In Air



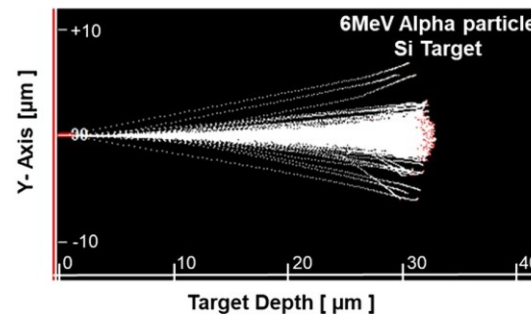
Air density
 1.293 kg m^{-3}

vs

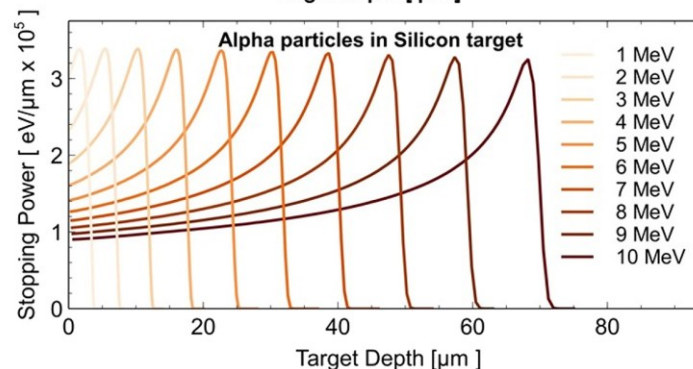
Silicon density
 2330 kg/m^3

In Silicon

(a)



(b)





Charge sharing effect - clusters

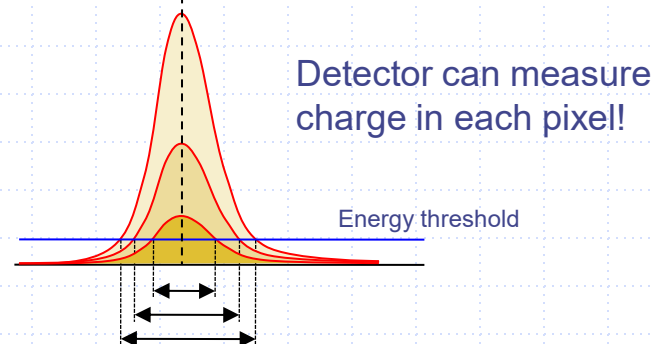
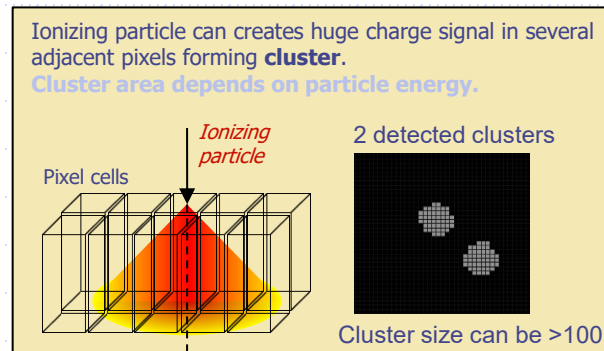


Cluster area brings analog information on energy deposited by a radiation quantum in the sensor

- ◆ Ionizing particle creates a charge in the sensor.
- ◆ The charge is collected by external electric field => the process takes some time
- ◆ Due to charge **diffusion** the charge cloud **expands**
- ◆ The charge cloud can overlap several adjacent pixels => **CLUSTER**
- ◆ Pixels in a cluster will detect the charge if it is higher than certain threshold

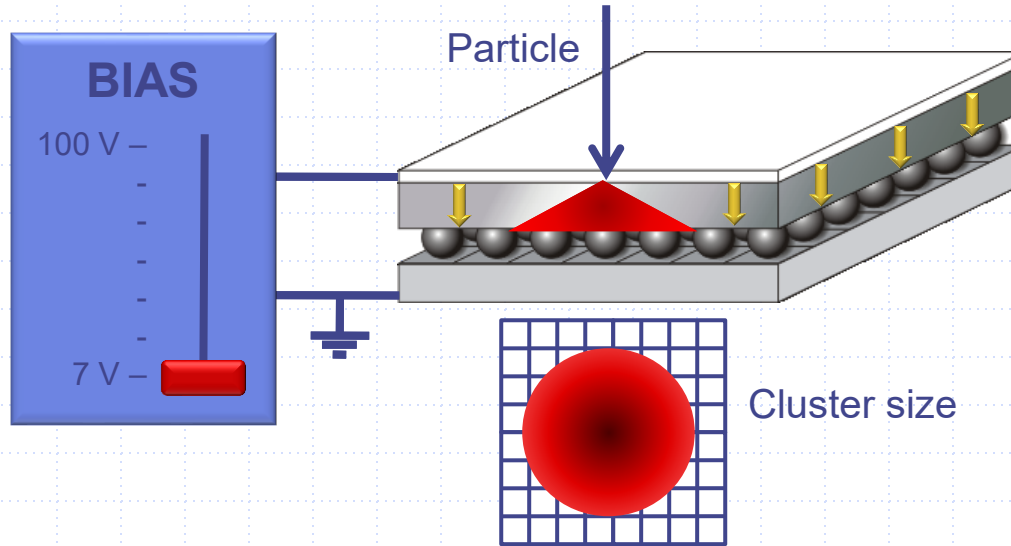
The Cluster size depends on:

- ◆ Particle energy and range
- ◆ Depth of interaction
- ◆ Detector Bias Voltage
- ◆ Local CCE (e.g. due to a material inhomogeneities and radiation damage)





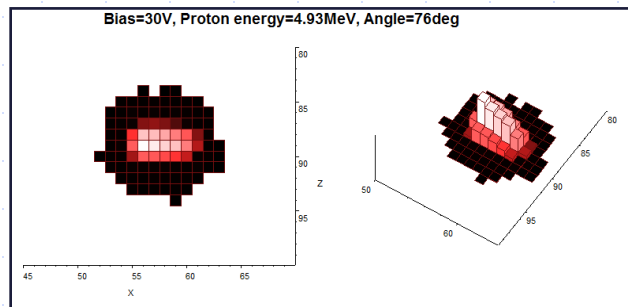
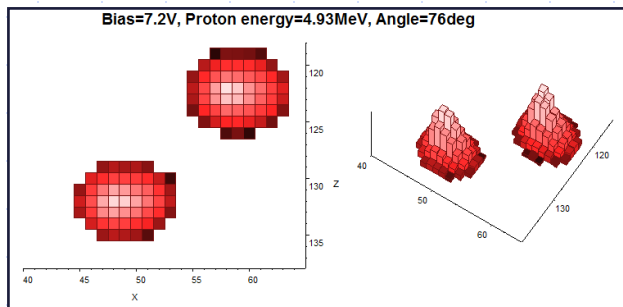
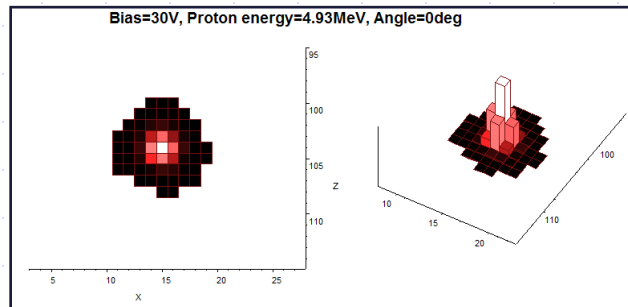
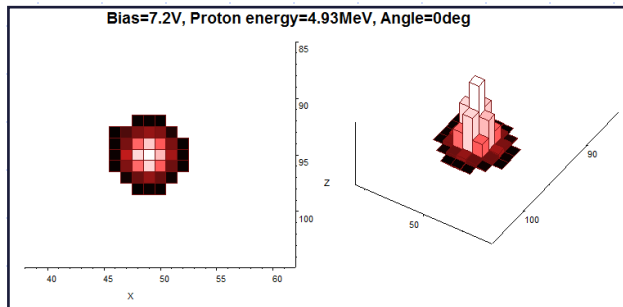
Dependence of cluster size on applied bias (on electric field in semiconductor)





4.93 MeV proton tracks recorded by Timepix silicon detector in Time-over-Threshold mode.

Illumination under different angles and different applied detector biases.



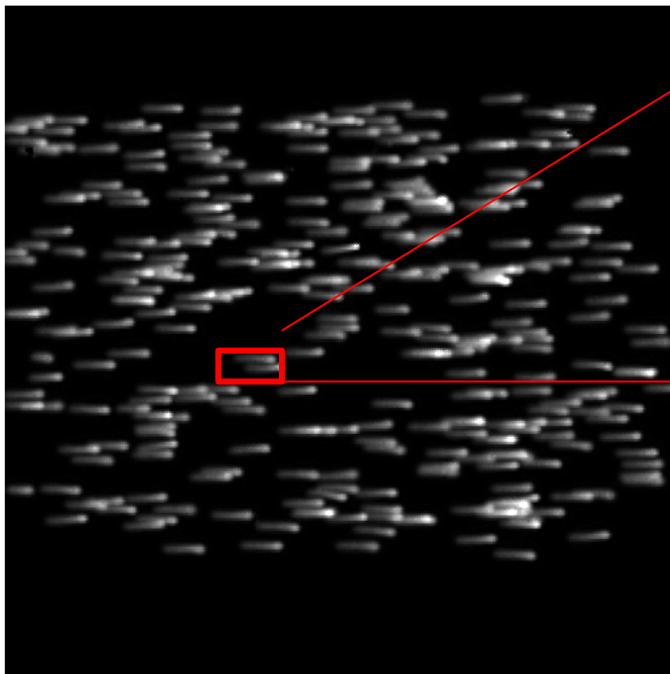
What is the spatial resolution? X- and Y-coordinates are determined with a precision of about 500nm. Determination of angle is with a precision of about 1°.



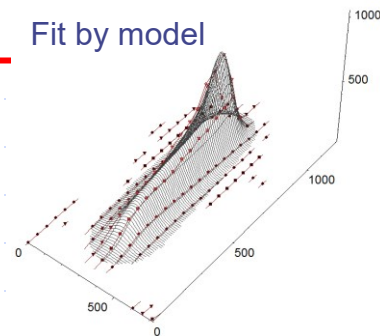
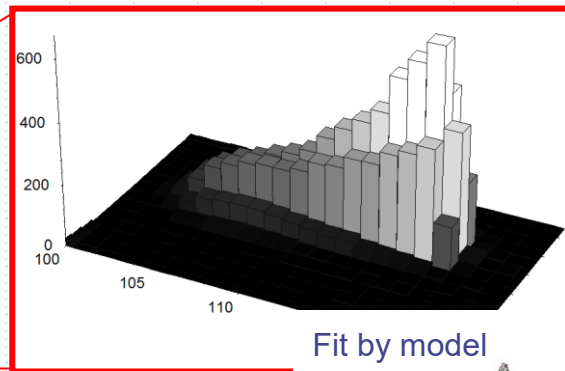
Flock of 11 MeV protons entering the silicon sensor under 85°



11 MeV protons, 85 degrees

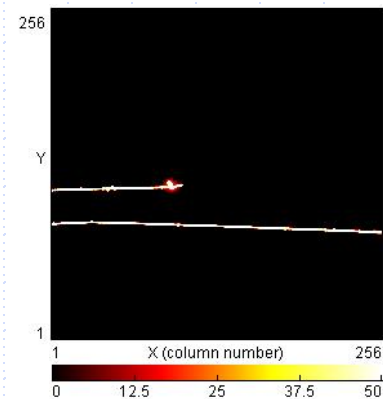
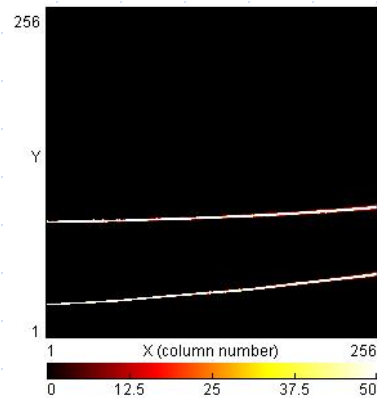
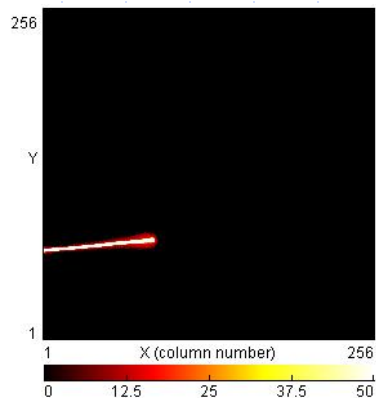
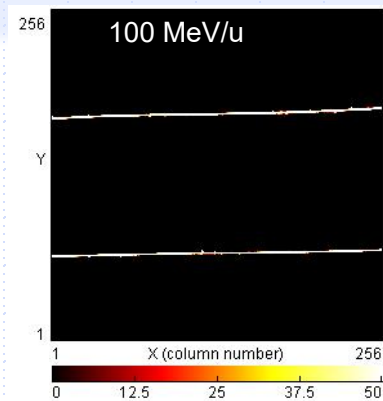
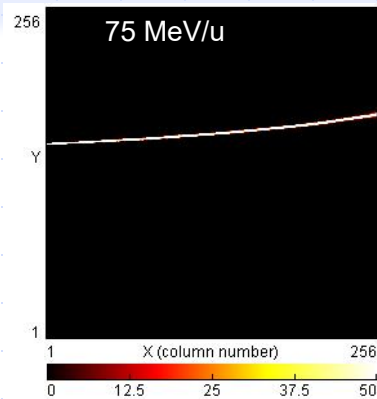
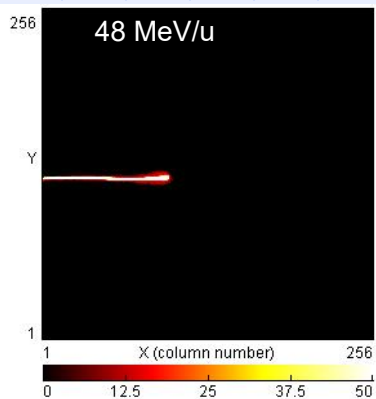


$\Delta E/\Delta x$ Bragg profile nicely pronounced, proton range about 960 μm



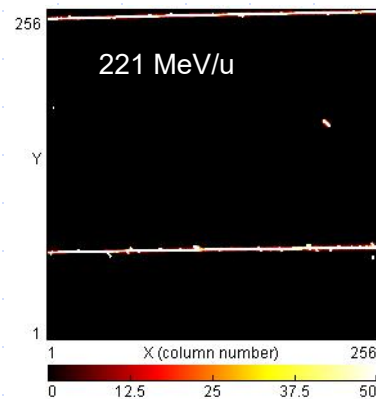
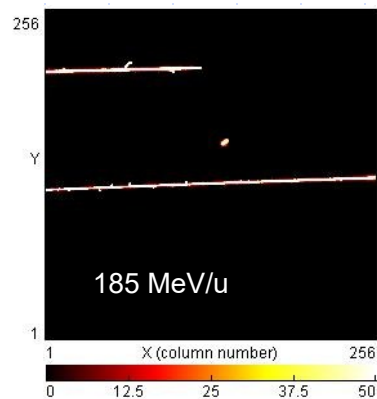
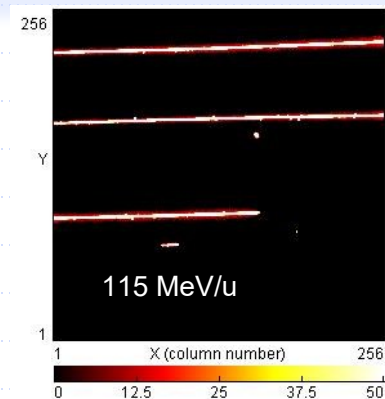
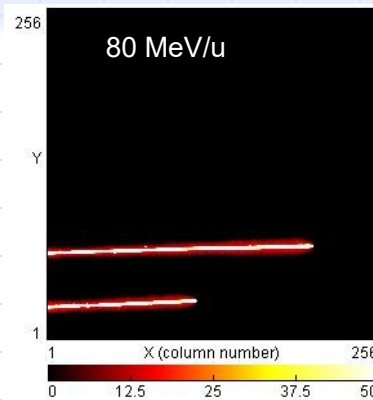
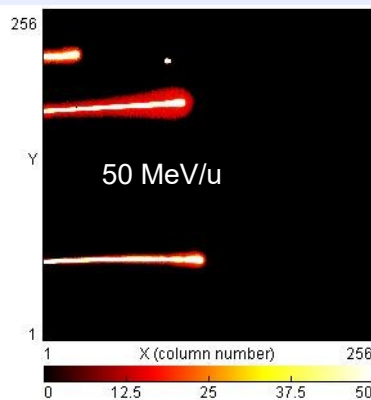


Protons with different energies at 90 degree



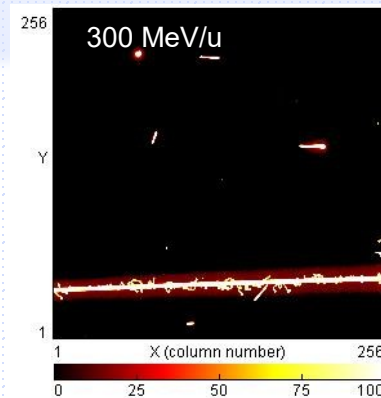
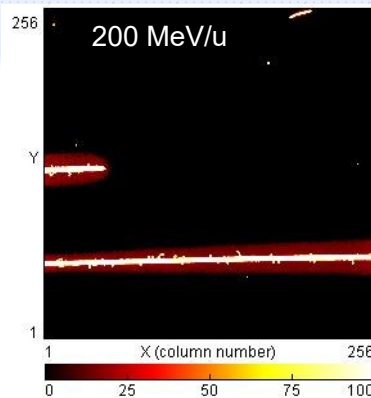
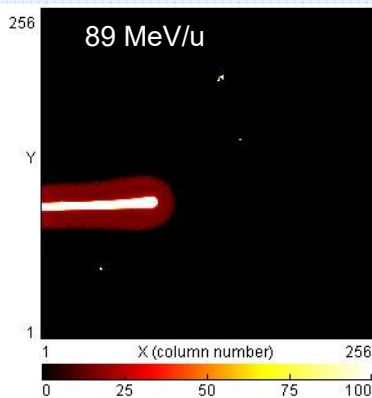
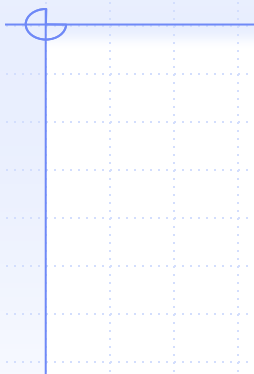


Alphas with different energies at 90 degrees

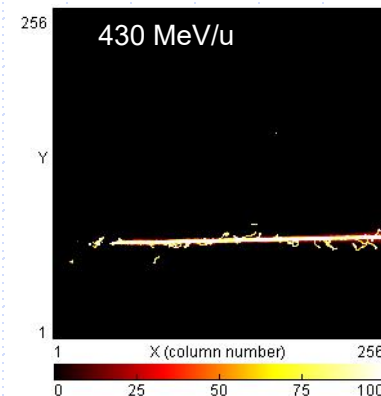




Carbon ions with different energies: 90 degrees

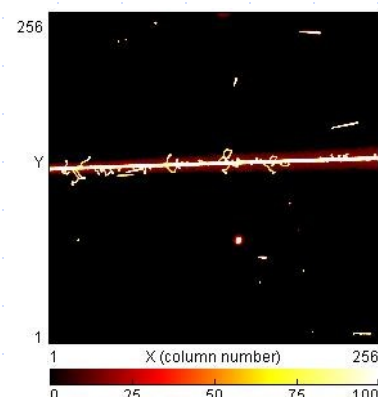
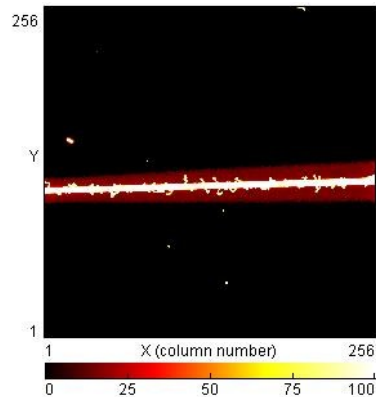
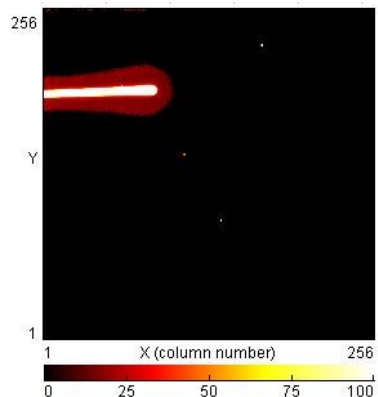
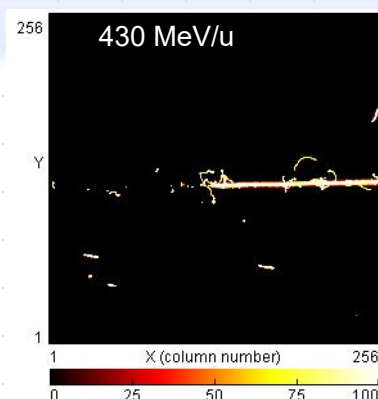
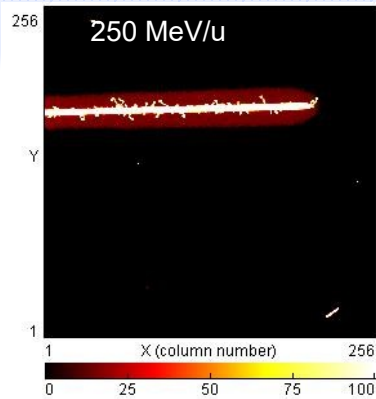
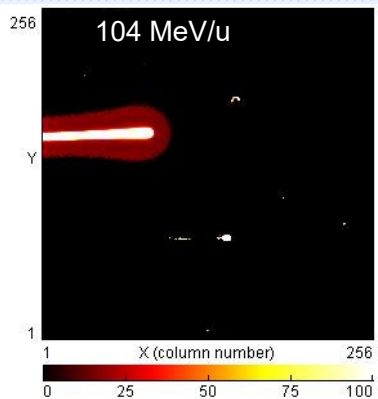


- ❑ Halo of pixel with low energy deposition around track - less pronounced for higher energies.
- ❑ Number of delta rays increases with increasing energy.





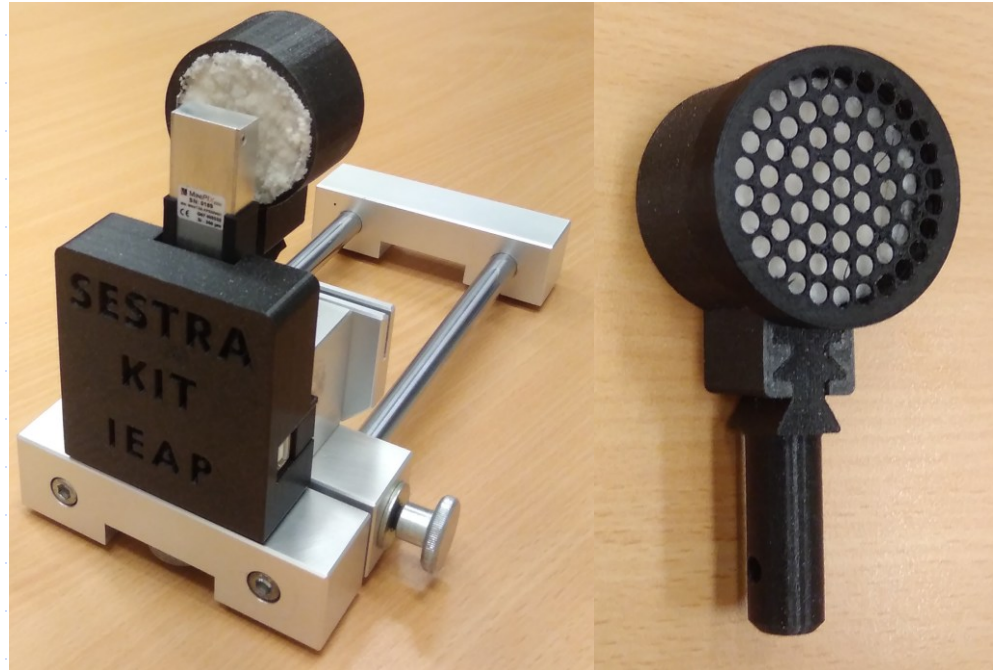
Oxygen ions with different energies: 90 degrees





Live Demonstration

Timepix detector exposed to Beta / Electrons





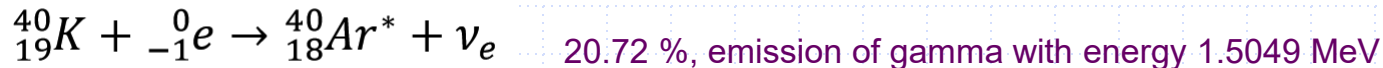
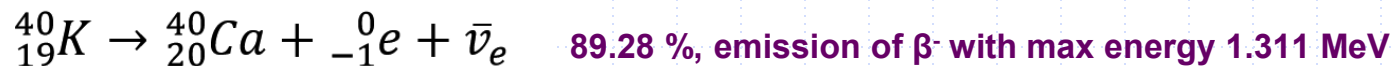
POTASSIUM Chloride Salt



- KCl - Potassium Chloride
- Natural abundance 0.0117% of radioactive isotope ^{40}K with half life $1.277 \cdot 10^9$ years

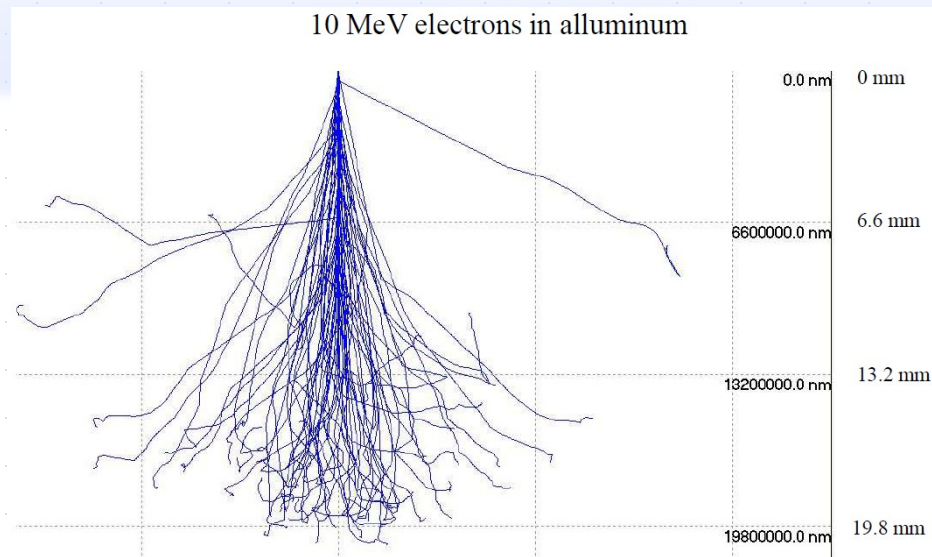
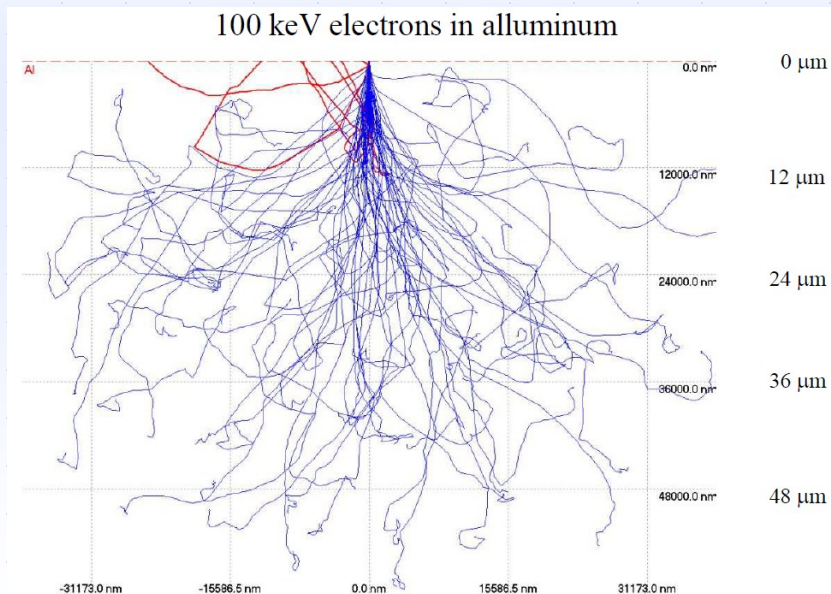


Decay modes





Electron trajectory - scattering



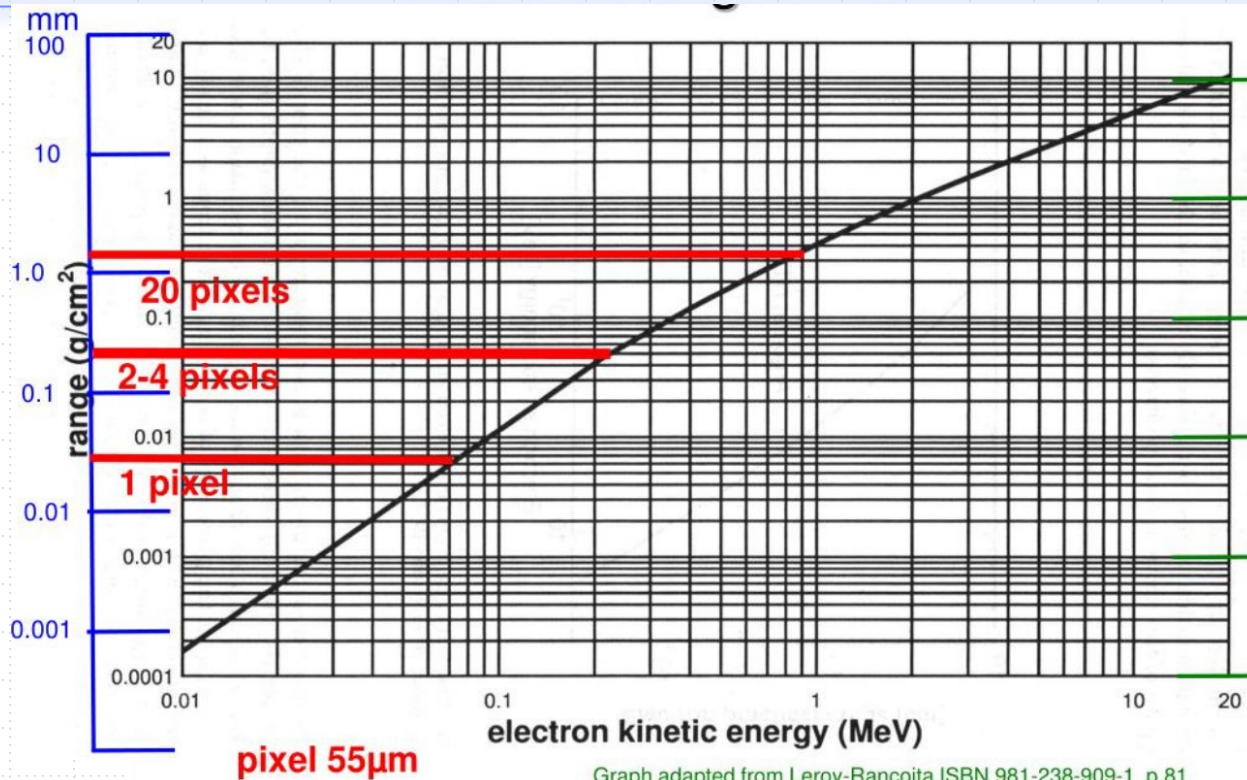
Monte Carlo simulation - CASINO code

- Probability of scattering by a large angle is significantly greater than for a heavy charged particle
- During a single collision, the electron can lose significantly more energy (up to half of the original energy)
- Fast electrons interacting with electrons in material (projectiles have same mass as targets)

Note significant difference in trajectory between low vs high energetic electrons
energetic electrons (with speed comparable to c) become relatively heavier



Electron range in silicon

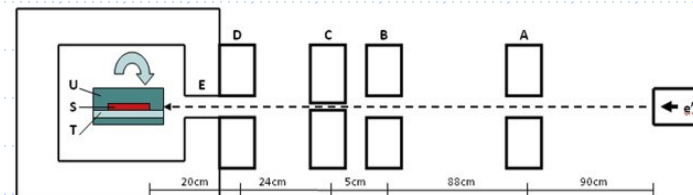
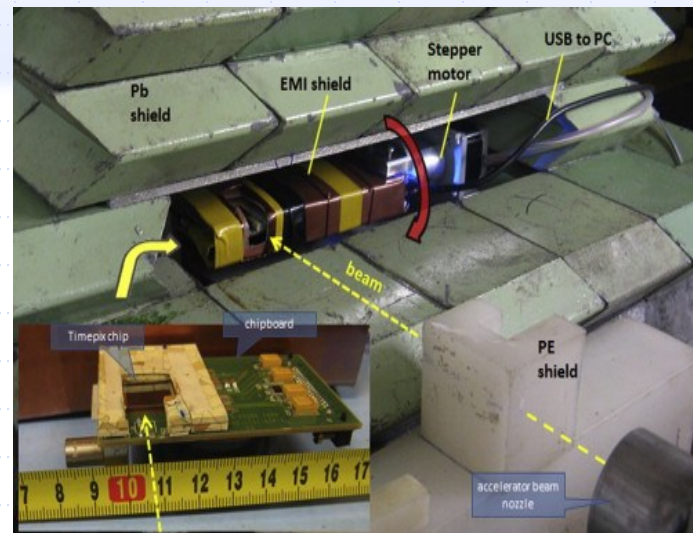
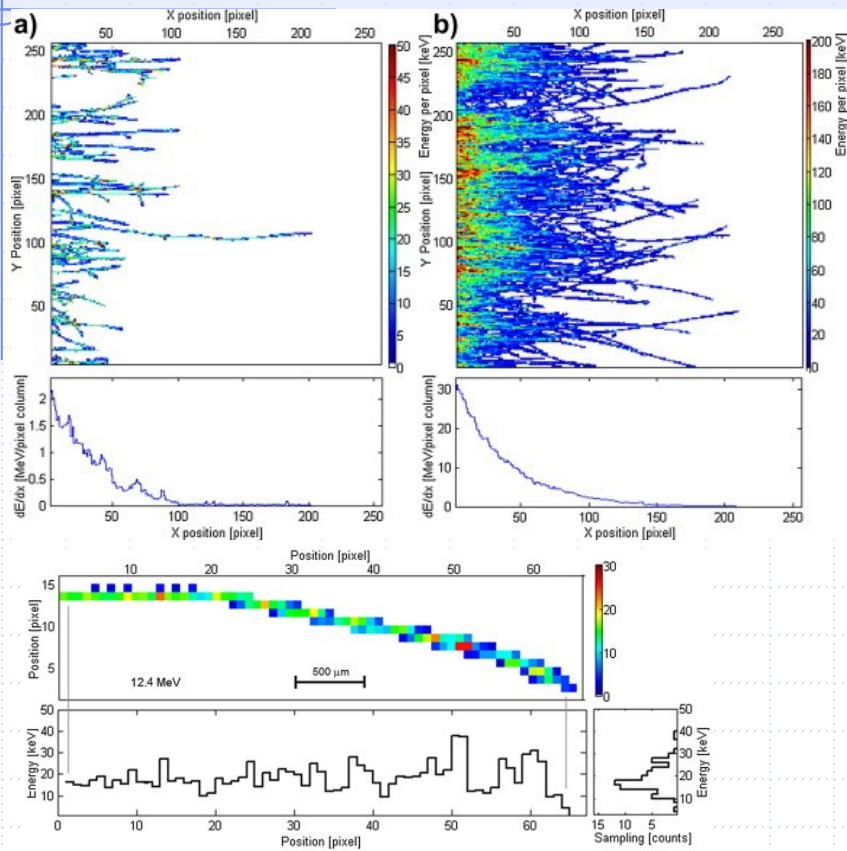


Graph adapted from Leroy-Rancoita ISBN 981-238-909-1 p.81



Observation of energetic electrons

Parallel beam and monoenergetic MeV electrons from the MT 25 microtron accelerator



Carlos et al., „Energy loss and online directional track visualization of fast electrons with the pixel detector Timepix“ <https://doi.org/10.1016/j.radmeas.2013.07.006>

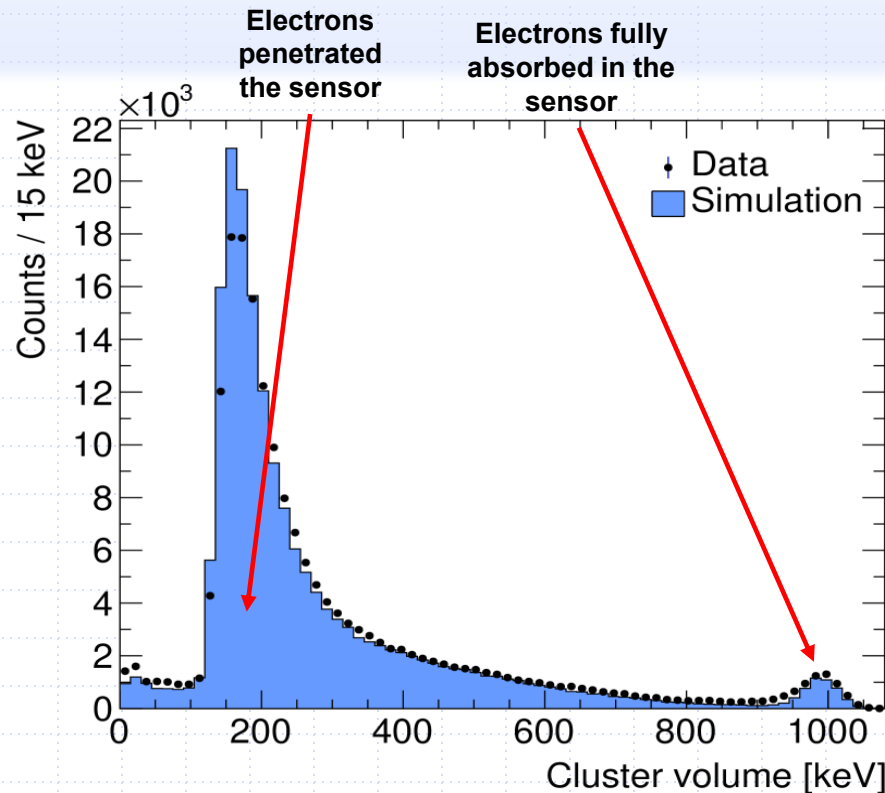


Observing electrons impacting Timepix



- Timepix3 + 500um thick Si sensor
- Detector exposed to 1MeV electrons
- 1MeV electrons have the range well exceeding the sensor thickness

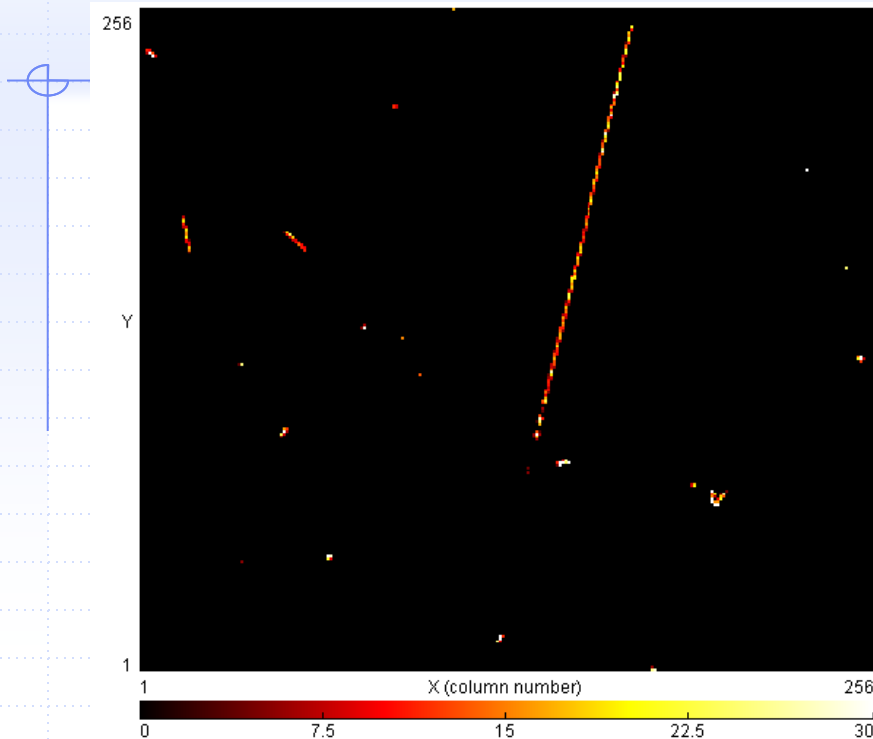
- *Energies (i.e., of the order of 1 MeV) are frequently multiply scattered within the detector sensor, leading to wiggly tracks which hinders determination of the particle incoming direction.*
- *On the other hand, it is these processes that allow full-energy deposition in a sensor with thickness smaller than would correspond to electron range calculated solely from the sensor-material linear stopping power*



Babar Ali et al, Study of electron tracks in Timepix3 detector at kinetic energies of 1 and 1.5 MeV



Observing Muons with Timepix detector



Review of muon properties

- Relativistic particles (their speed is comparable to c)
- Highly penetrative
- Rest mass of muon is about 200x heavier than electron

Typical
response
to muons
**Straight
track**



Timepix allows us to determine directionality of detected muon tracks

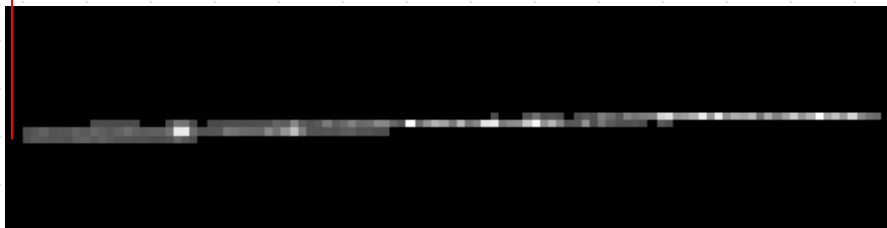
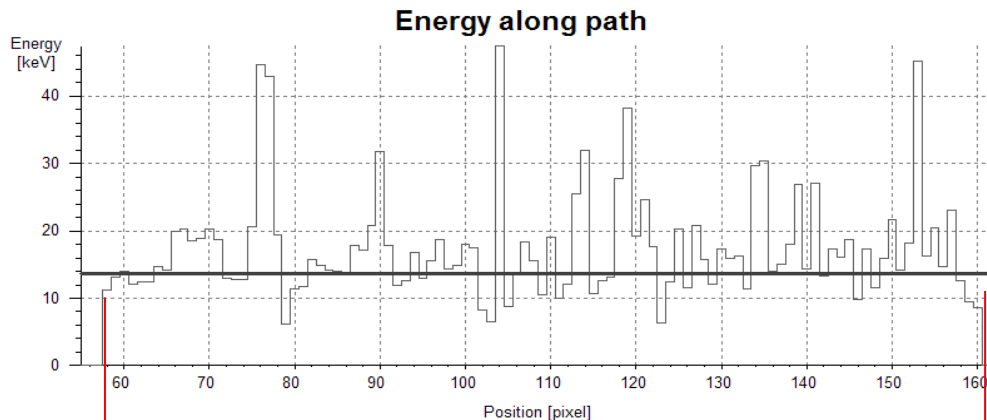
Note the unfavorable geometrical factor of sensor

- Timepix sensor is relatively flat typically 14mm x 14mm x 300um (or 500um)
- muons can be clearly recognized if they arrive almost in parallel to sensor surface



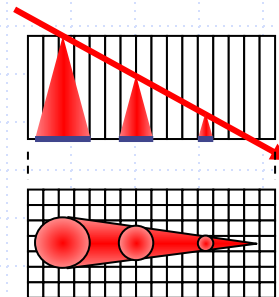
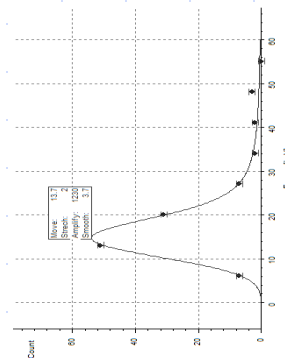
Track of MIP particle – Cosmic muon

Charge sharing effect helps to determine all track coordinates and direction with quite high resolution
(deeply submicrometric in case of x and y)



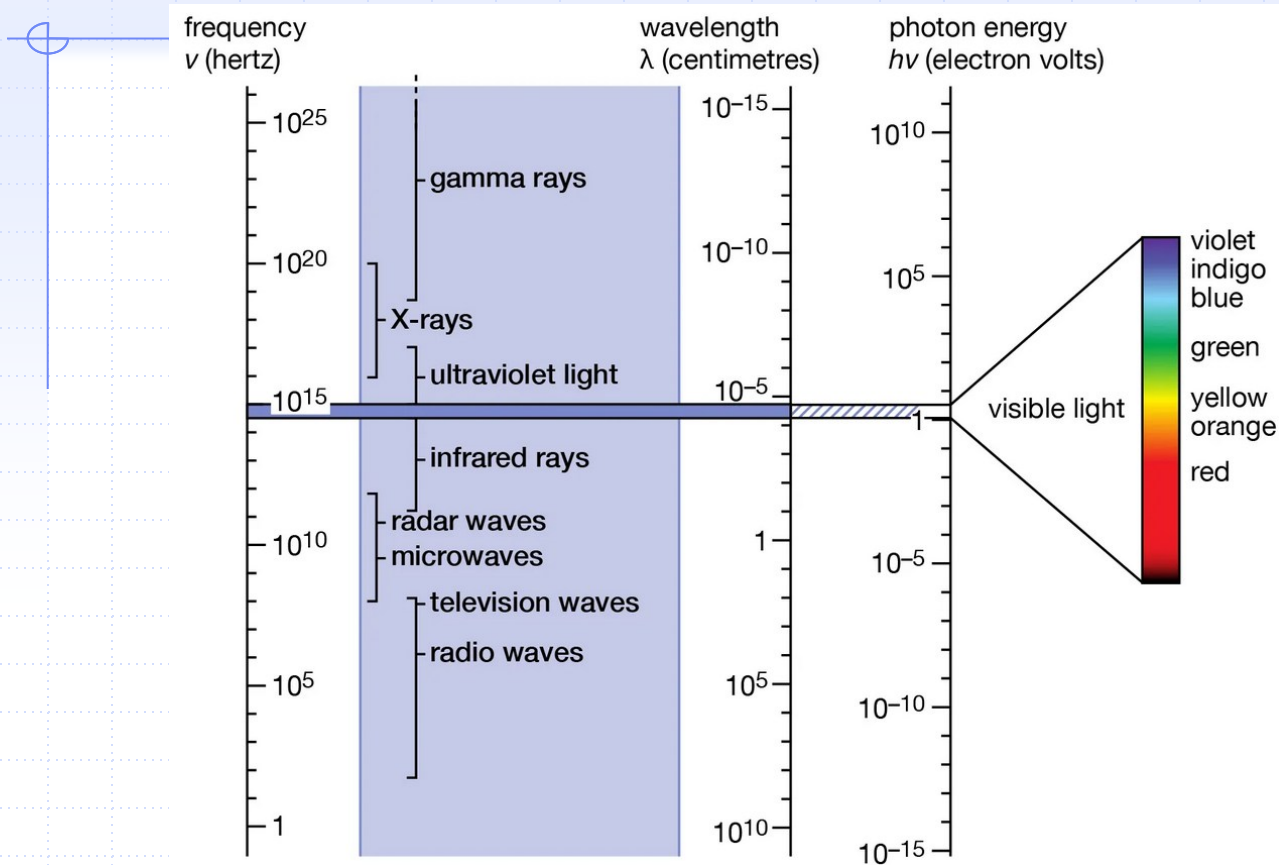
Muon track recorded by Timepix detector

Energy distribution fit by convolution of Landau and Gauss distribution





Detection of Gamma / X-ray photons



Note>

Timepix detector
pixel threshold energy
is about 2 to 3 keV



Detection of Gamma / X-ray photons

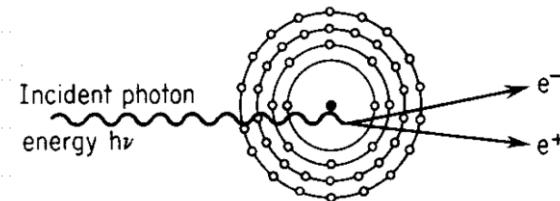
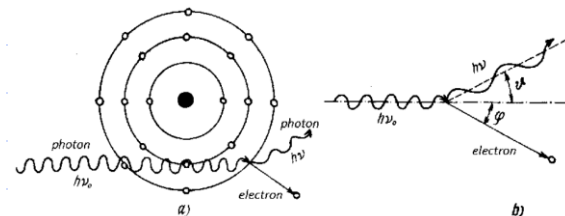
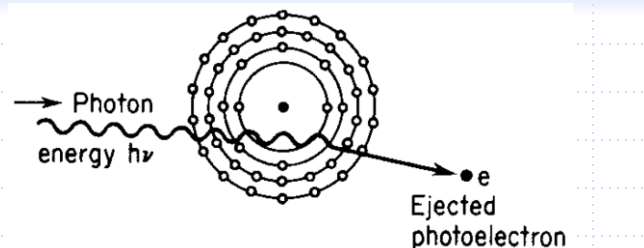
Indirectly ionizing radiation

Detector can see secondary radiation, i.e. electrons and positrons (these secondaries represent directly ionising particles)

Interaction in a sensor mainly via these processes:

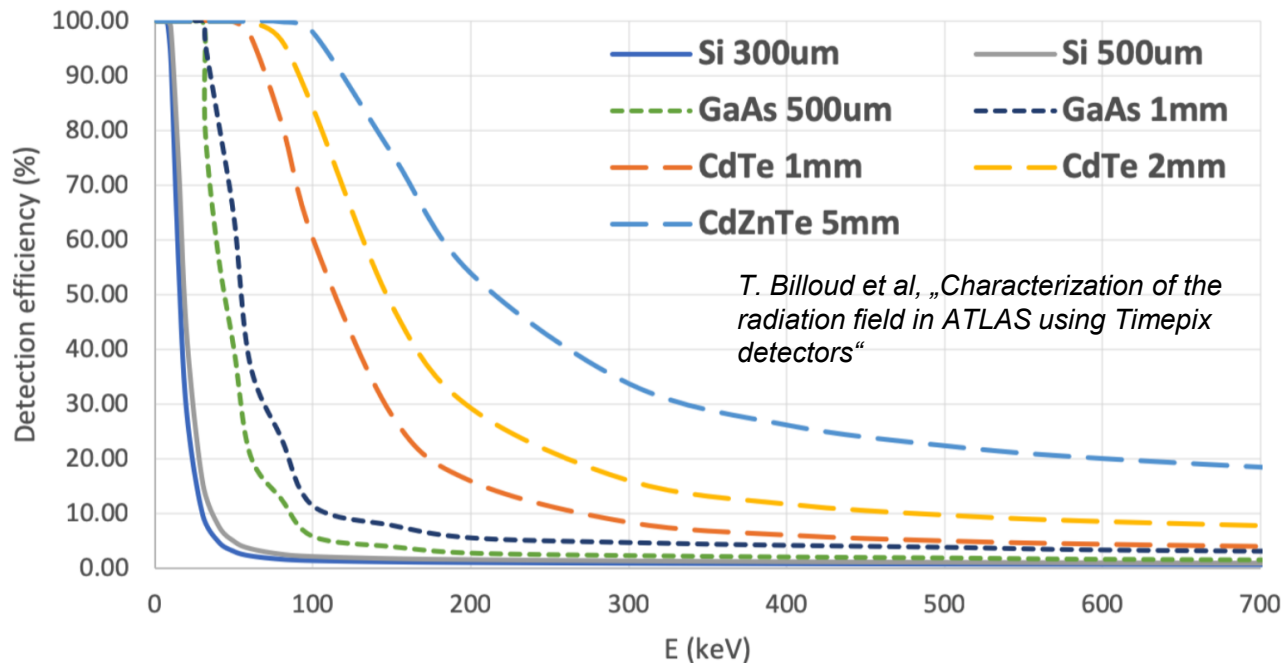
- Photoelectric effect
- Compton scattering
- Positron/Electron pair production

Note that X-Ray/Gama photons are finally displayed in the pixel matrix in the same way as betas/electrons





Detection efficiency of X-ray/Gamma photons in various materials and impact of sensor thickness

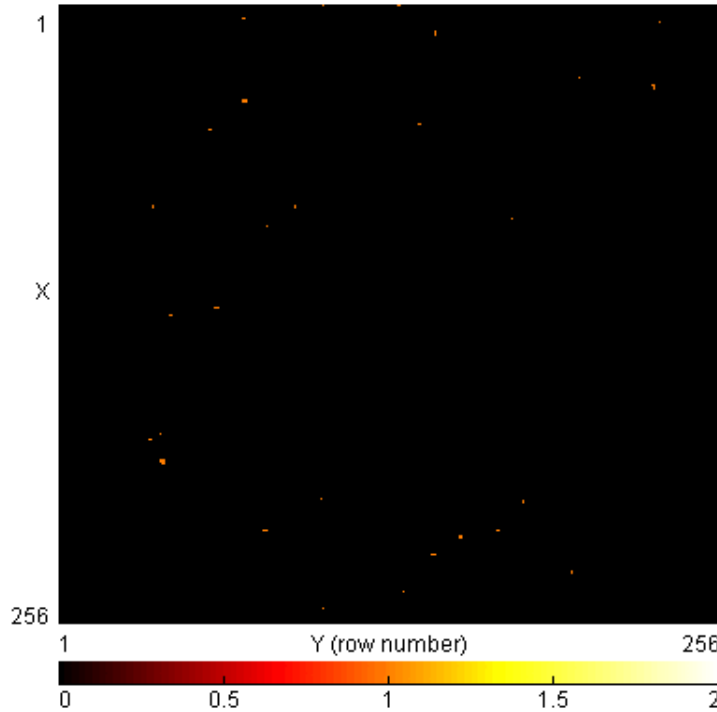


Note significant impact of sensor thickness and sensor density (i.e. sensor material) on detection efficiency in the case of energetic photons



(Xray) Radiography Demonstration

- Prove of basic working principle

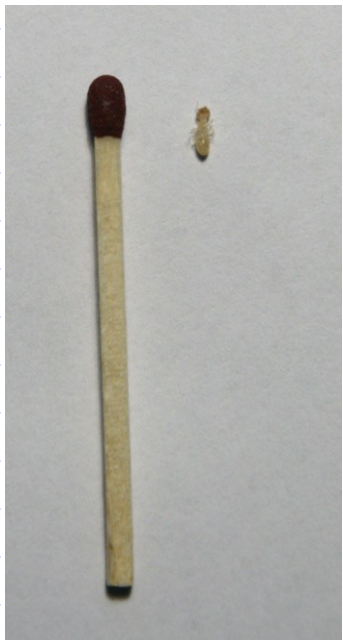


Metalic object is placed in between the radiation source and Timepix detector



High resolution X-ray radiography:

Imaging of Termites



The imaging of termites as a model **soft tissue organism** is particularly difficult due to their **poorly sclerotized** cuticle making difficult to observe the anatomic structures with an optimal contrast.

Moreover, they are vulnerable to damage when they are manipulated or treated during sample preparation.

Thus, the termites represent an ideal model to optimize the accuracy and sensitivity of the developed method.



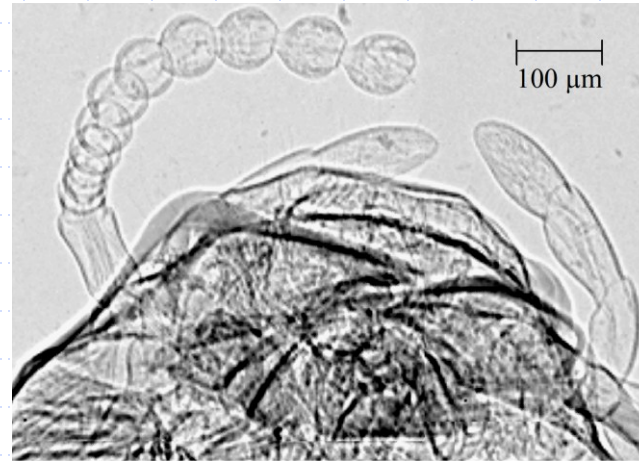
High resolution X-ray radiography:

Imaging of Termites



X-ray transmission image of termite worker body (left) and detail of its head (bottom). Even fine internal structure of the antennae is recognized.

(Magnified 15x, time=30s, tube at 40kV and 70 μ A)





High resolution X-ray radiography: Imaging of Living Termites



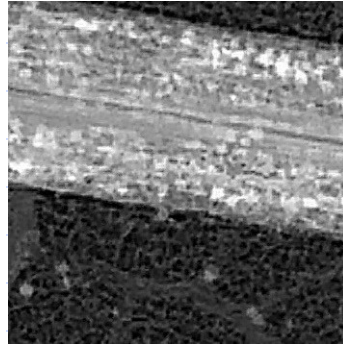
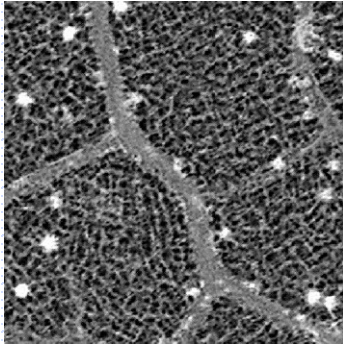
Images of a termite worker before (left) and after (right) its metamorphosis toward the soldier caste (5s exposure $\sim 0.7\text{mGy}$ dose)



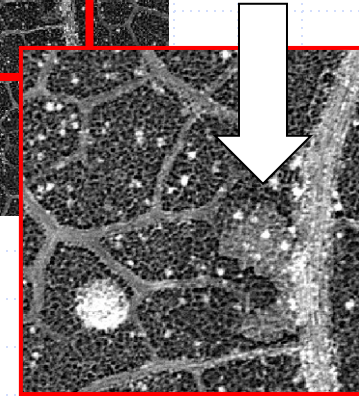
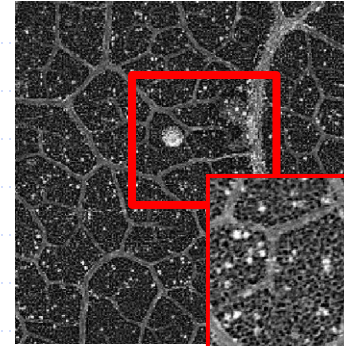
High resolution X-ray radiography: Leaf Miner story

Leaf miner (*Cameraria ohridella*) - small moth. In larvae stadium it lives inside of chestnut tree leafs making "mines" and causing serious problems to the tree. Indication: chestnut leafs get brown, dry and fall down early.

Courtesy of J.Dammer (CTU in Prague), P.M.Frallicciardi (U.of Napoli) and F. Weyda (SBU Ceske Budejovice)



Healthy chestnut tree leaf structure (no parasite) – cellular structure of leaf is nicely observed (resolution below 1 μm). The white spots are small drops of resin secreted by the leaf.

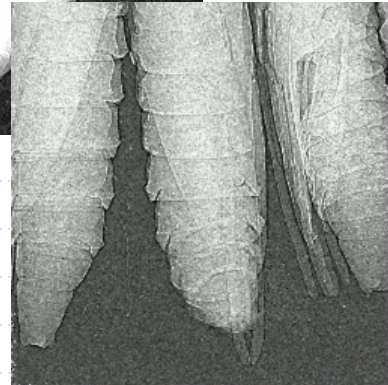
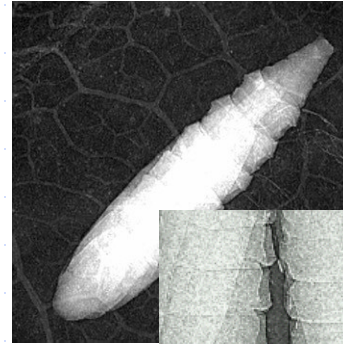
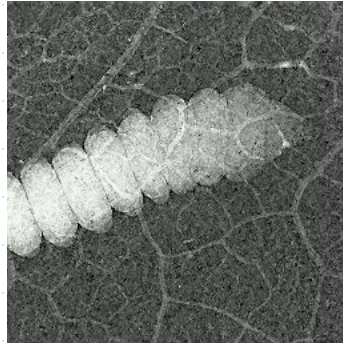
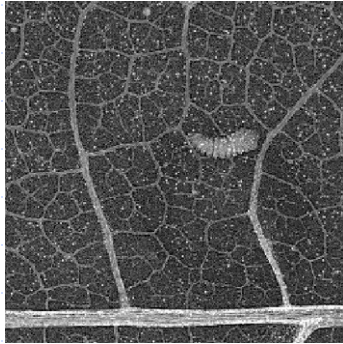




High resolution X-ray radiography:

Example: Leaf Miner story

Worms are growing up and after three feeding instars larvae build-up a silken cocoon (pupae)

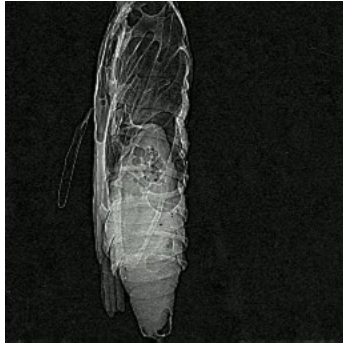


Several
collected
pupas



High resolution X-ray radiography: Example: Leaf Miner story - Cure

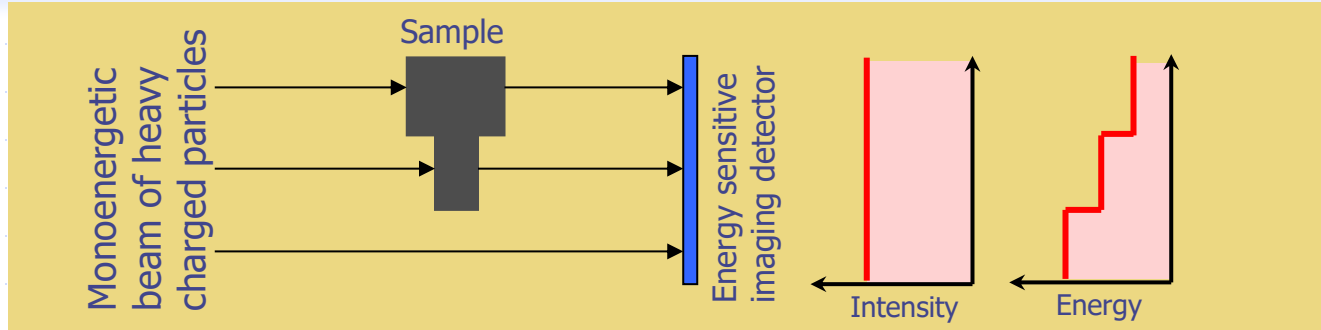
- ◆ The best cure: natural enemy (parasitic wasp)
- ◆ Certain small wasps can put eggs into leaf miner pupa
- ◆ Parasite inside of parasite:



Parasite kills the
pupa and leaves it
as adult wasp



Radiography with highly ionizing particles



- ◆ **Heavy charged particles** (protons, deuterons, tritons, alphas, ions) can be used (impossible with photons, difficult with electrons due to huge change of direction).
- ◆ Instead of transmitted beam intensity the **energy losses** of individual particles are measured.
- ◆ Just a single particle is needed to measure material "density".
- ◆ With common sources of heavy charged particles (isotopes, ion beams) it is feasible to inspect small objects (thin layers, foils, cellular structures)
- ◆ **Precision of thickness measurement can be in nanometer scale**



Simple alpha radiography with Timepix pixel detectors



By cluster analysis it can be determined:

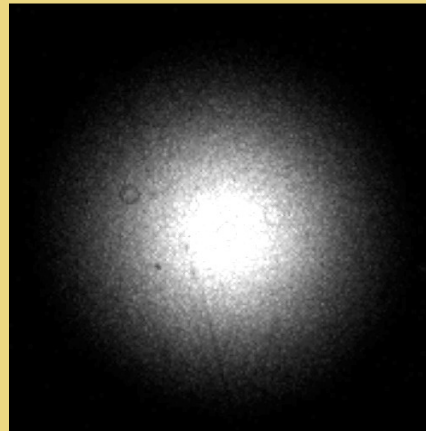
- **Centroid** to increase spatial resolution (subpixel resolution)
- **Size** as a measure of particle energy

Wing of fly

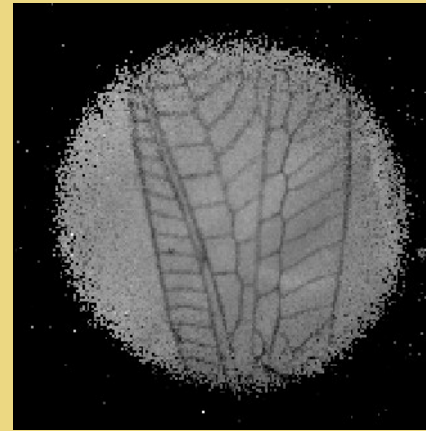
(less than 20 particles per pixel)



Photo



Intensity



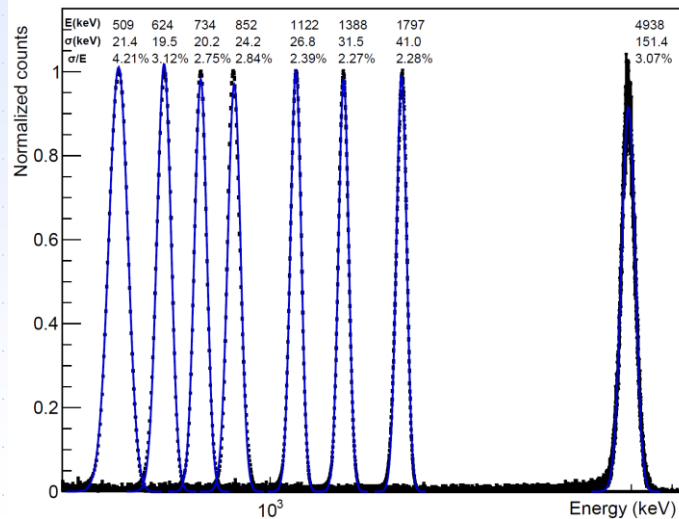
Energy



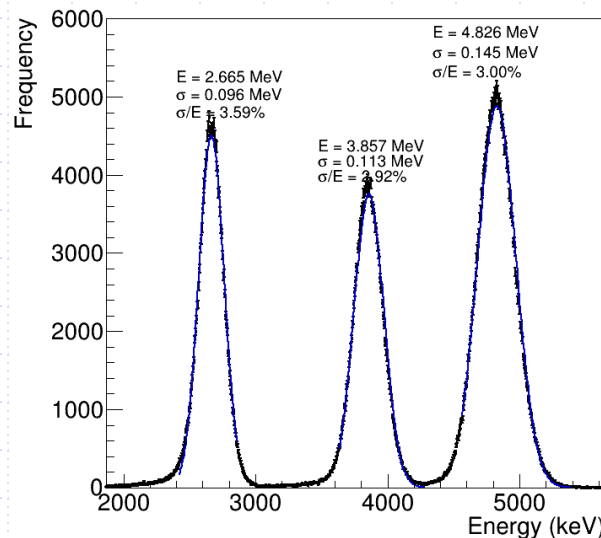
Working principle



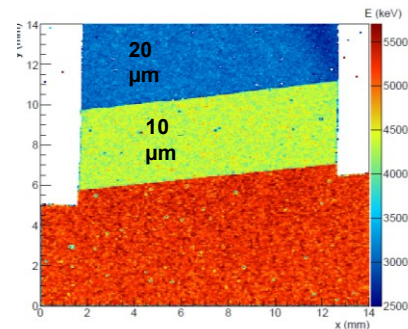
Proton and alpha spectra



For stopped protons <3% relative resolution



For short-ranged alpha particles <4% relative resolution



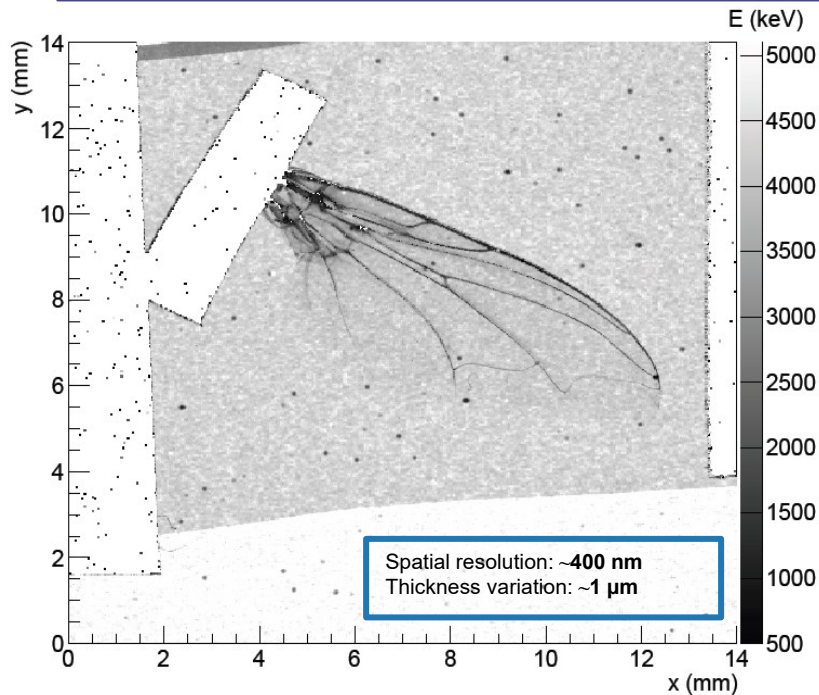
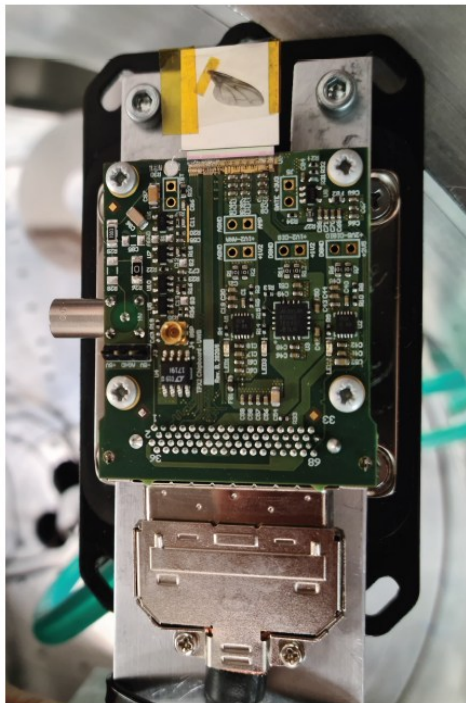
Mylar foils attached to the sensor backside



Alpha particle radiography



Alpha source placed at 6 cm distance from the detector, measured in vacuum



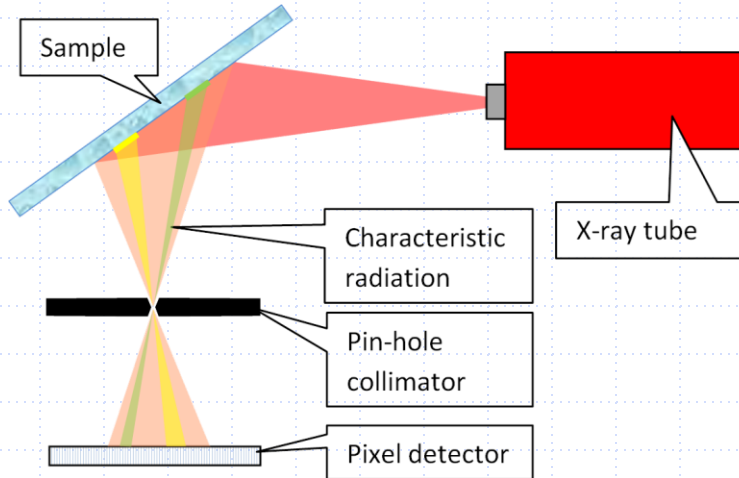
B. Bergmann et al 2022 JINST 17 C01025



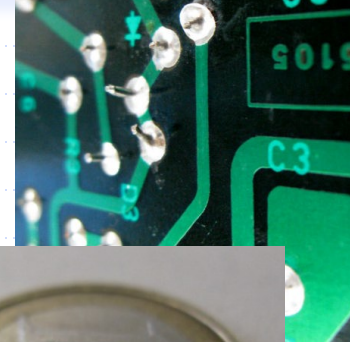
Recognition of materials

Detection of X-ray fluorescence photons

Experimental setup and samples



Piece of PCB



One Euro coin

- Sample of unknown composition is exposed to X-ray radiation
- Fluorescence photons are emitted from the sample
- Spectrum of fluorescence photons is analyzed



Characteristic energy of X-Ray fluorescence photons for each element



Periodic Table of Elements and X-ray Energies

www.bruker.com/hhxf
sales.hmp@bruker.com

Note

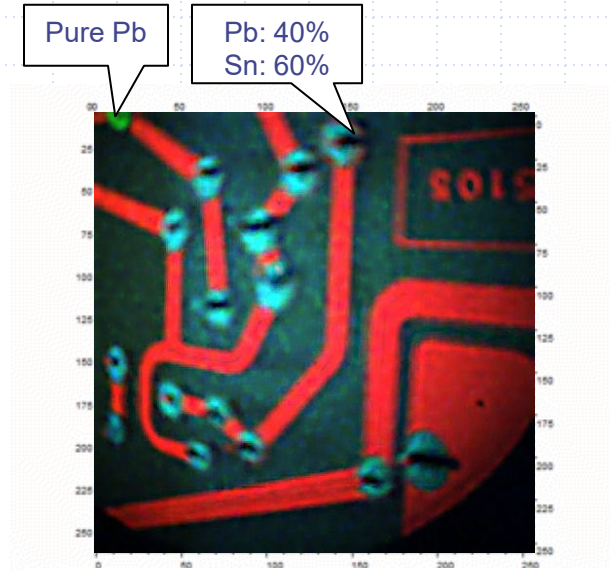
- Higher energies for high-Z elements
- Energy of primary X-ray photons must be higher than energy of fluorescence photons

Z	Element	K α_1	K β_1	L α_1	L β_1
22	Ti Titanium	4.512	4.933	0.452	0.458
23	V Vanadium	4.953	5.428	0.510	0.518
24	Cr Chromium	5.415	5.947	0.572	0.582
25	Mn Manganese	5.900	6.492	0.637	0.648
26	Fe Iron	6.405	7.059	0.705	0.718
27	Co Cobalt	6.931	7.649	0.775	0.790
28	Ni Nickel	7.480	8.267	0.849	0.866
29	Cu Copper	8.046	8.904	0.928	0.947
30	Zn Zinc	8.637	9.570	1.012	1.035
31	Ga Gallium	9.251	10.267	1.098	1.125
32	Ge Germanium	9.886	10.982	1.188	1.218
33	As Arsenic	10.543	11.726	1.282	1.317
34	Se Selenium	11.224	12.497	1.379	1.419
35	Br Bromine	11.924	13.292	1.481	1.526
36	Kr Krypton	12.648	14.112	1.585	1.636
37	Rb Rubidium	13.396	14.961	1.692	1.751
38	Sr Strontium	14.165	15.835	1.806	1.871
39	Y Yttrium	14.958	16.739	1.924	1.998
40	Zr Zirconium	15.775	17.668	2.044	2.126
41	Nb Niobium	16.615	18.625	2.169	2.260
42	Mo Molybdenum	17.480	19.606	2.292	2.394
43	Tc Technetium	18.367	20.626	2.423	2.535
44	Ru Ruthenium	19.279	21.656	2.558	2.683
45	Rh Rhodium	20.216	22.724	2.697	2.834
46	Pd Palladium	21.177	23.818	2.838	2.990
47	Ag Silver	22.163	24.941	2.983	3.150
48	Cd Cadmium	23.173	26.093	3.133	3.315
49	In Indium	24.210	27.275	3.286	3.487
50	Sn Tin	25.271	28.485	3.444	3.663

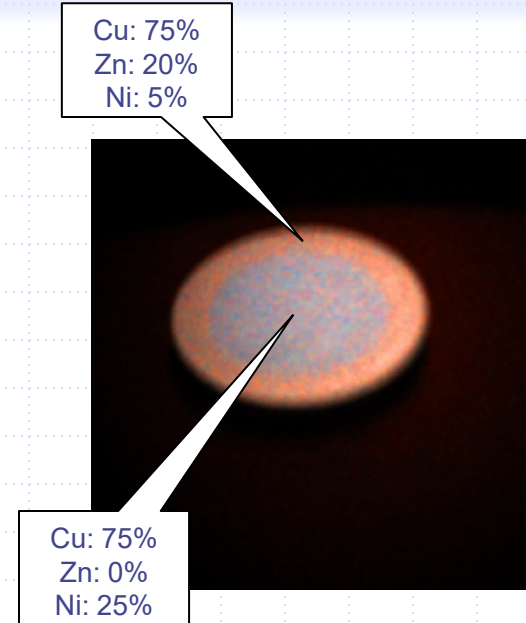


X-ray fluorescences

Results of exemplar observation



Color coding:
Cu = Red, **Pb** = Green, **Sn** = Blue



Color coding:
Zn content is displayed in Pink

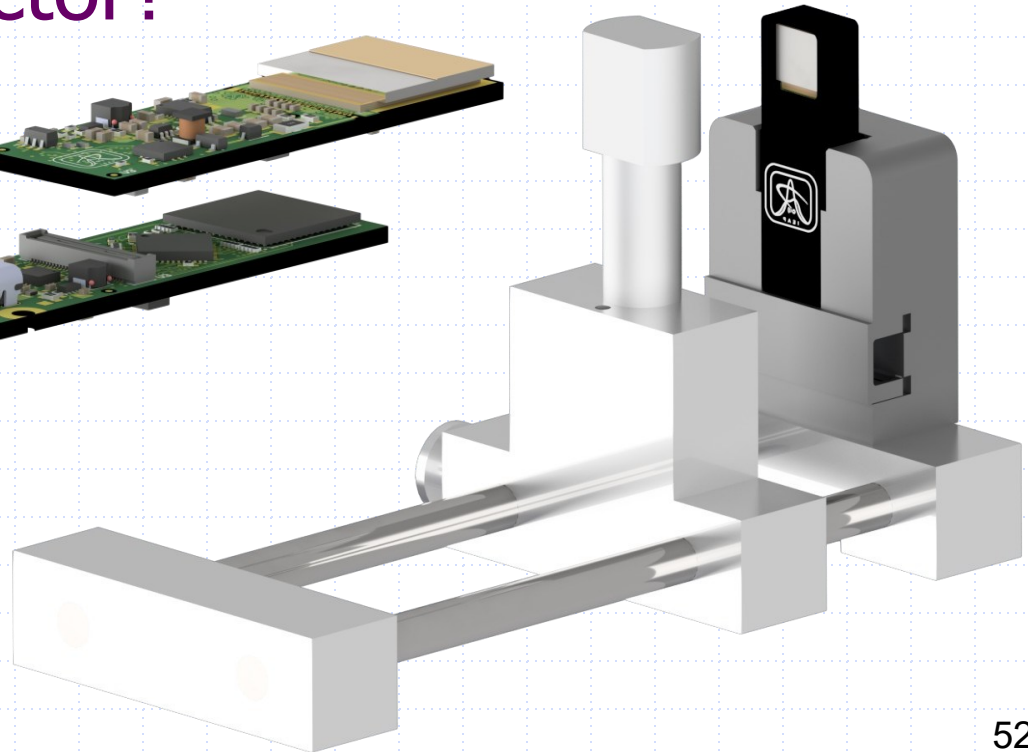
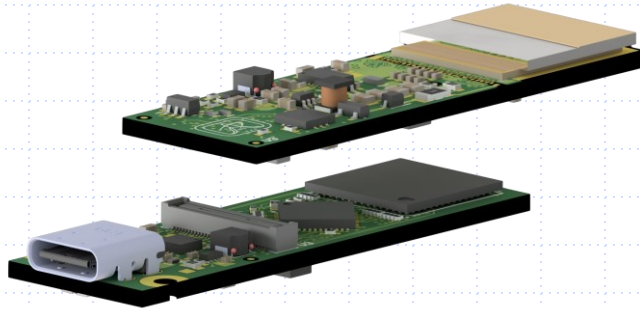
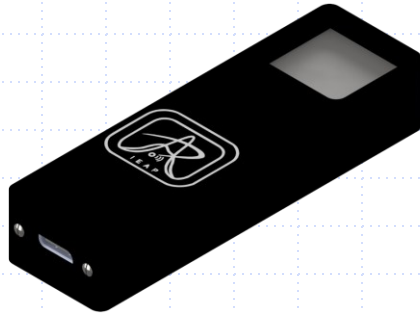


What if we know when exactly particles hit the detector?

...exploiting Time of Arrival (ToA) measurement mode



How we can measure half life of the second excited state of ^{237}Np with Timepix2 detector?



Alfa decay - Americium:



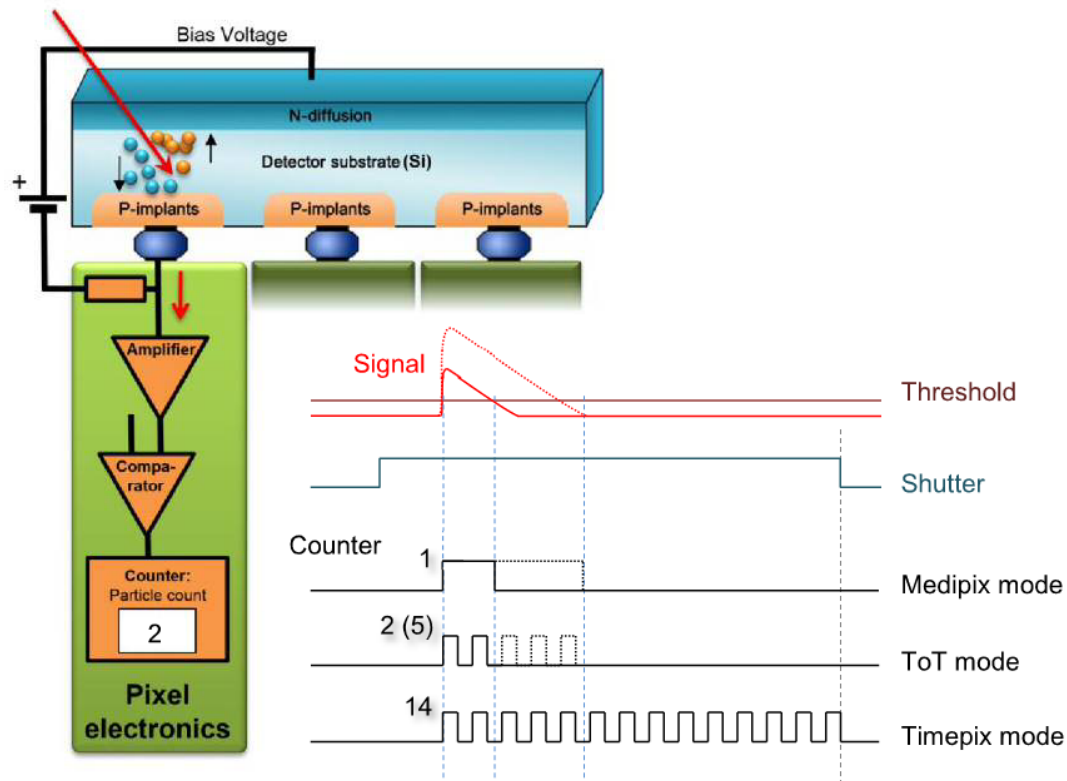
$$E(\alpha) = 5\,500\text{ keV}$$

+ Gamma emitted after Np deexcitation

$$E(\gamma) = 59.5\text{ keV}$$



Timepix2 - Pixel operation modes

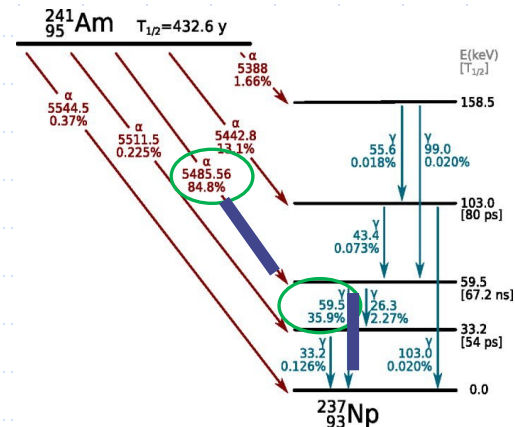


Example of pixel measurement modes:
ToA18/ToT10
 ➔ 18 bits ToA, 10 bits ToT information
 (energy measured in parallel to arrival time)

Mode Name	Description	1 st Counter	2 nd Counter
ToT10/ToA18	Simultaneous ToT and 1 st hit ToA (with sequential read/write)	10-bit ToT {CounterA}	18-bit ToA {CounterD, CounterC, CounterB}
ToT14/ToA14	Mode options (programmable): 1) 1 st hit or integral ToT 2) Overflow (wraparound) of ToA counter	14-bit ToT {CounterD, CounterA}	14-bit ToA {CounterC, CounterB}
ContToT10/Event4	Continuous read/write ToT Mode options: 1) 1 st hit or integral ToT (programmable) 2) Supplementary 4-bit eventing counting (readout optional)	10-bit ToT {CounterA} 4-bit #Events {CounterD}	10-bit ToT {CounterB} 4-bit #Events {CounterC}
ContToT14	Continuous read/write ToT Mode option: 1 st hit or integral ToT (programmable)	14-bit ToT {CounterD, CounterA}	14-bit ToT {CounterC, CounterB}
ContToA10	Continuous read/write 1 st hit ToA	10-bit ToA {CounterA}	10-bit ToA {CounterB}
ContToA14		14-bit ToA {CounterD, CounterA}	14-bit ToA {CounterC, CounterB}
ContEvent10	Continuous read/write event counting	10-bit #Events {CounterA}	10-bit #Events {CounterB}
ContEvent14		14-bit #Events {CounterD, CounterA}	14-bit #Events {CounterC, CounterB}

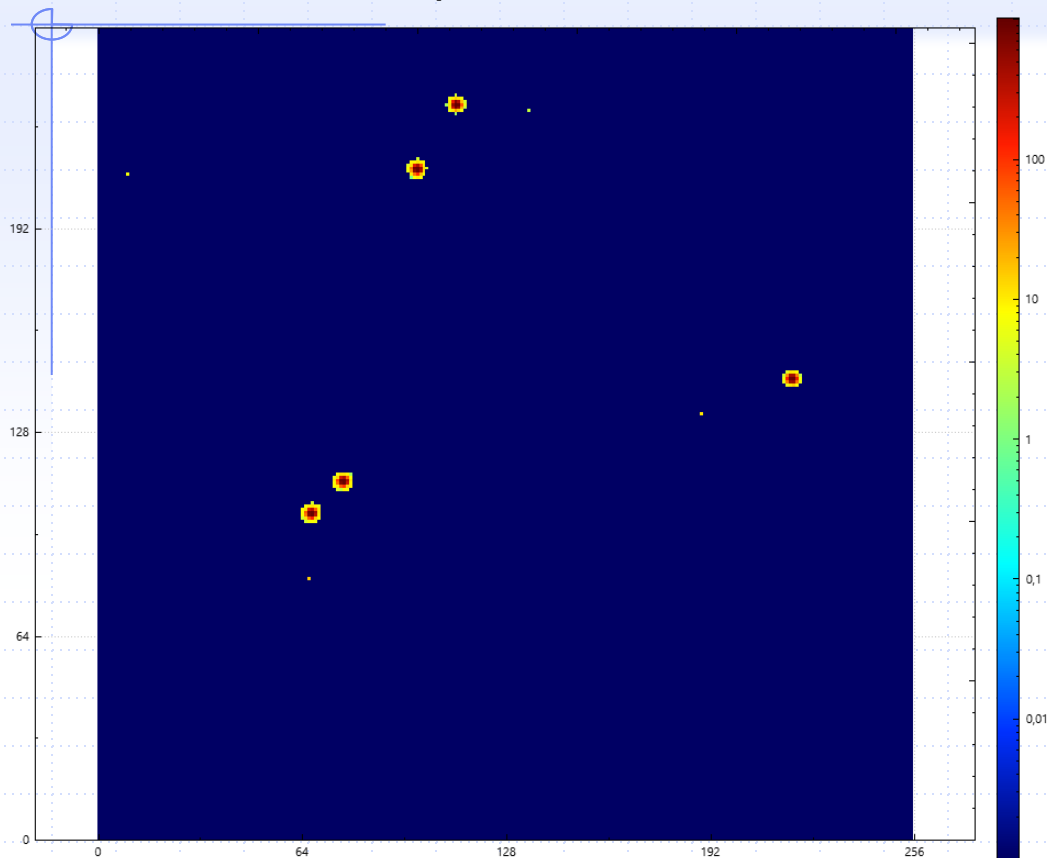


Zoom into the decay scheme of ^{241}Am (all energies are in keV)



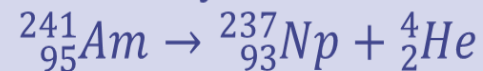


Exemplar frame when we observe ^{241}Am with TPX2



Events of interest

Alfa decay - Americium:



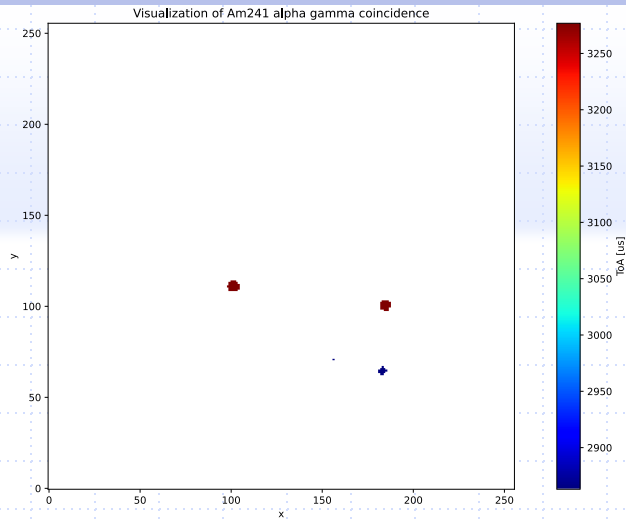
$$E(\alpha) = 5\,500\text{ keV}$$

+ Gamma emitted after Np deexcitation

$$E(\gamma) = 59.5\text{keV}$$

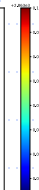
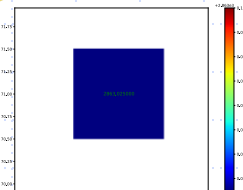
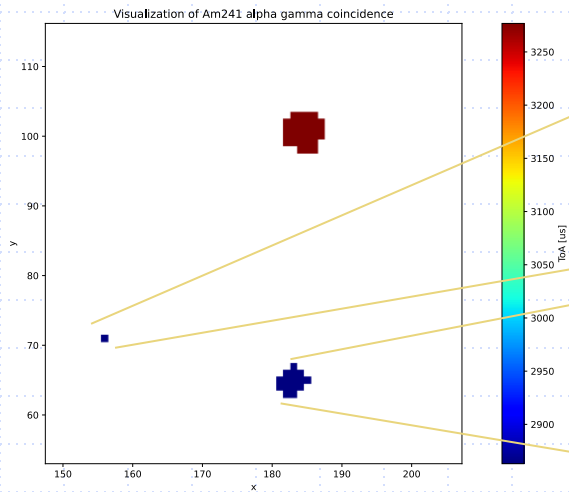


Close look and data analysis



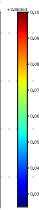
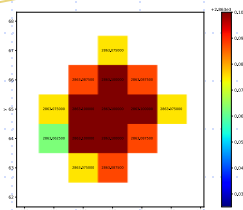
..Zoom..

One frame – ²⁴¹Am measurement



$$\text{ToA}_\gamma = 2863.025 \text{ us}$$

Time difference:
75 ns

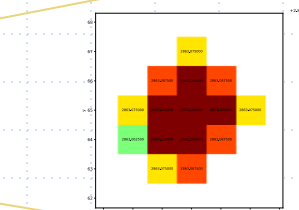
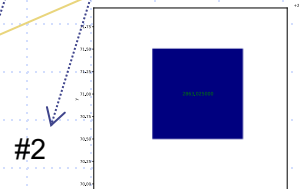
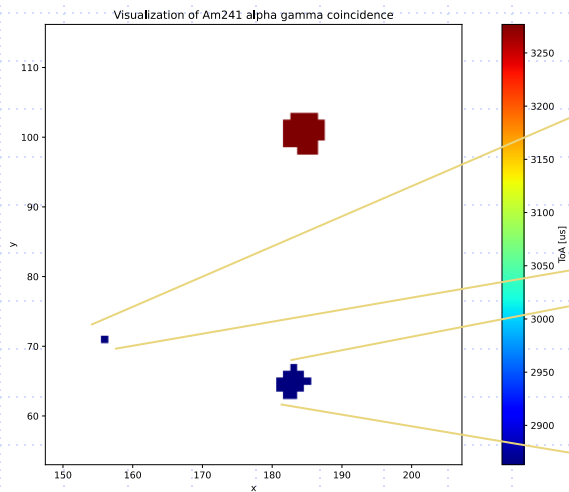
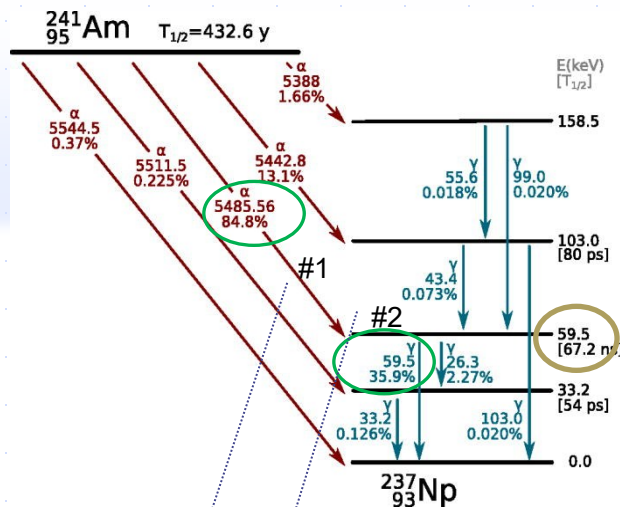


$$\text{ToA}_\alpha = 2863.100 \text{ us}$$

56



Close look and data analysis



$$\text{ToA}_\gamma = 2863.025 \text{ us}$$

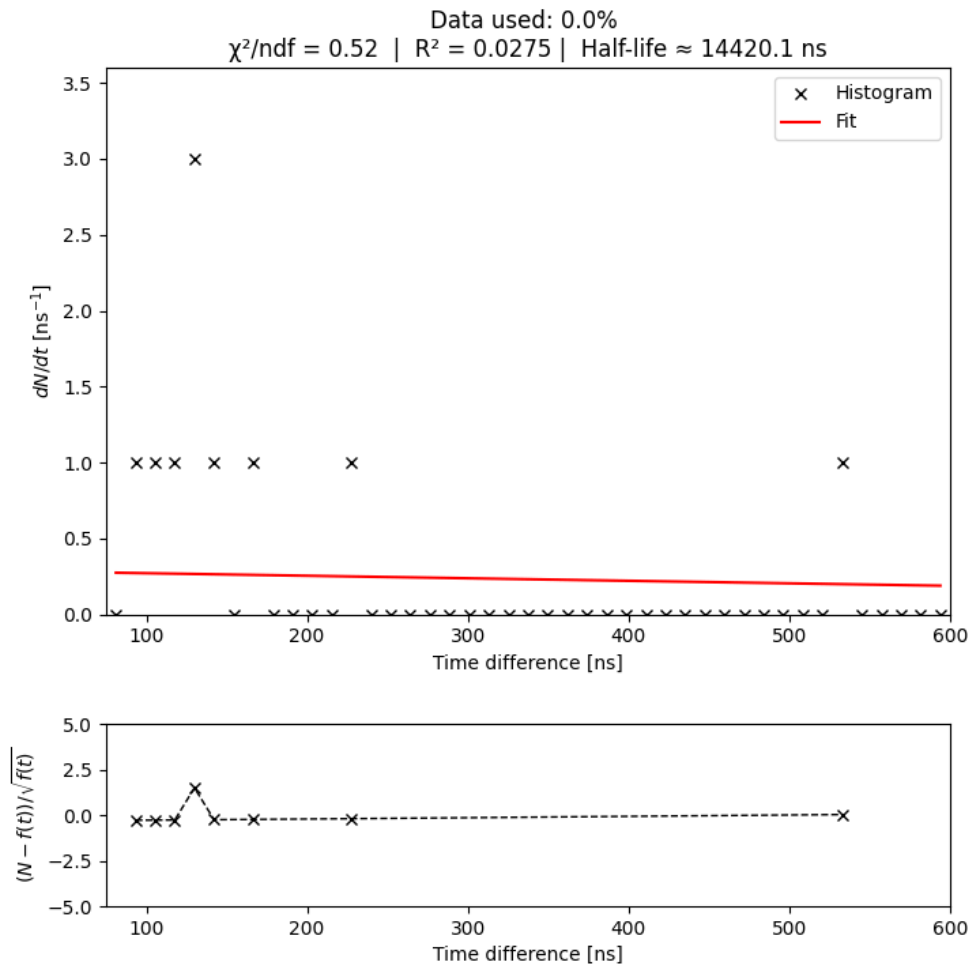
Time difference:
75 ns

$$\text{ToA}_\alpha = 2863.100 \text{ us}$$



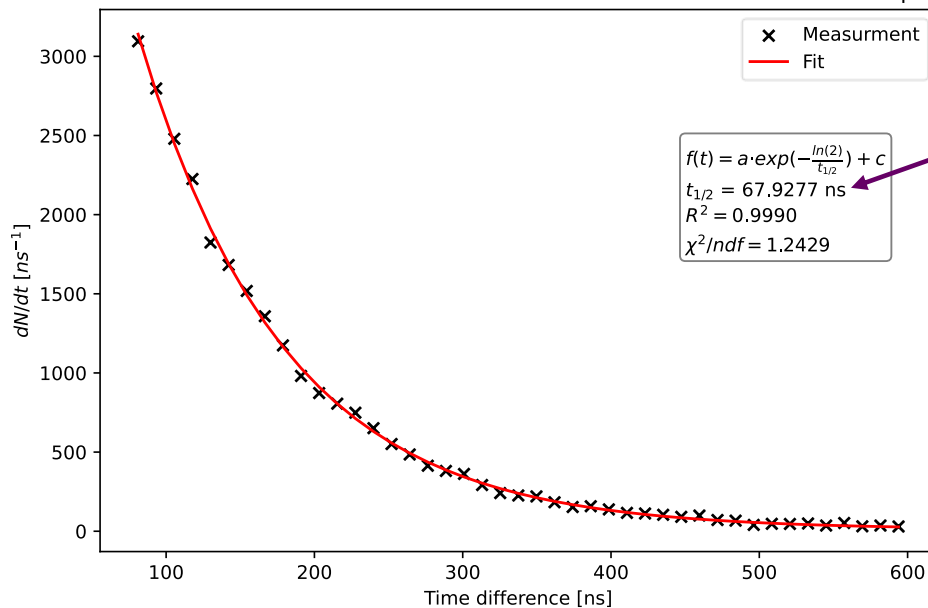
Measuring half life of the second excited state of ^{237}Np

Collecting more and more events...





The measurement of the half-life of the second excited state of ^{237}Np



Final Result

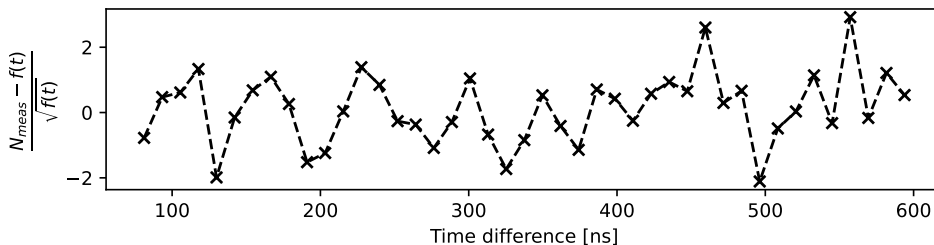
...after processing one day long measurement

What is the half life of second excited state of ^{237}Np ?

Values reported in other publications:

[2] : 68.1 ns

[1]: 67.7 ns



- [1] Vretenar, D. et al, (2019), *Introduction to covariant density functional theory for nuclear structure*. **American Journal of Physics**, 87, 997–1006. <https://doi.org/10.1119/1.5126823>
- [2] Basunia, M. S. (2006). *Nuclear Data Sheets for A = 237*. **Nuclear Data Sheets**, 107(8), 2323–2422. <https://doi.org/10.1016/j.nds.2006.07.001>



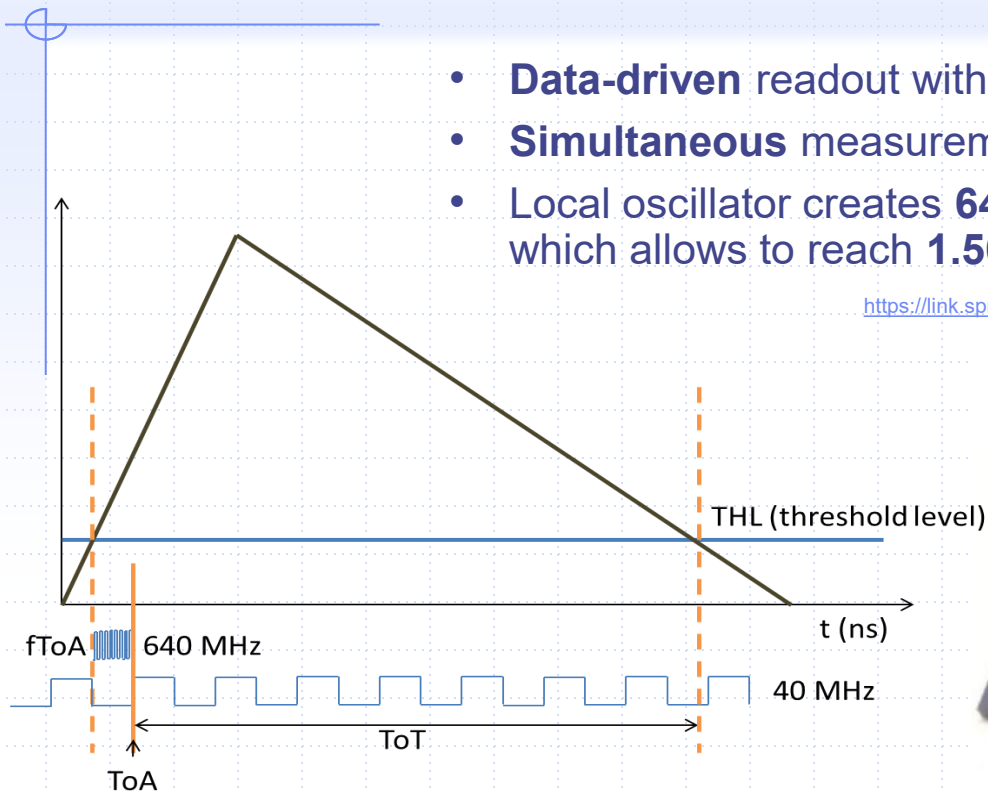
Working principle

Timepix3 – TOA (Time of Arrival)



- **Data-driven** readout with continuously running 40 MHz base clock
- **Simultaneous** measurement of ToT and ToA
- Local oscillator creates **640 MHz** clock for precise time stamping which allows to reach **1.56 ns** resolution of **ToA**

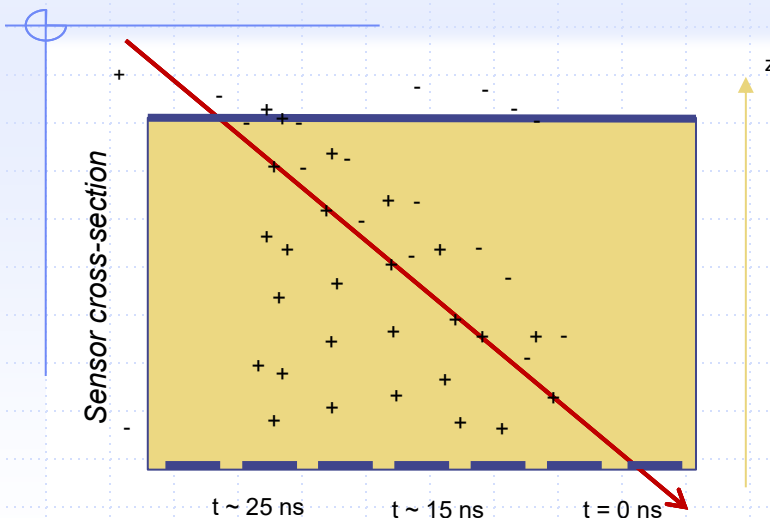
<https://link.springer.com/article/10.1140/epjc/s10052-017-4993-4>



Timepix3 with chipboard



3D reconstruction of particle tracks



- Propagation of deposited charge to the pixel electrode takes time
- High resolution of ToA measurement allows to observe time differences between pixels

Charge carrier drift motion:

e^- and h^+ drift described by

$$v_e = -\mu_e \times E(z)$$

$$v_h = \mu_h \times E(z)$$

$\mu_{e/h}$: Mobility of e^-/h^+

Electric field parametrization:

$$\text{Si: } \vec{E}(z) = \frac{U_B}{d} \vec{e}_z + \frac{2U_{dep}}{d^2} \left(\frac{d}{2} - z \right) \vec{e}_z;$$

$$\text{CdTe: } \vec{E}(z) = \frac{U_B}{d} \vec{e}_z$$

U_B : Bias voltage; U_{dep} : Depletion voltage; d : Sensor thickness

→ Look-up table: $z(t_{meas}, E_{meas.})$

Exemplar typical values of charge mobility in silicon

μ_h mobility of holes $450 \text{ cm}^2/\text{V}\cdot\text{s}$

μ_e mobility of electrons $1350 \text{ cm}^2/\text{V}\cdot\text{s}$

Bergmann et al. Eur. Phys. J. C (2017) 77: 421. <https://doi.org/10.1140/epjc/s10052-017-4993-4>

Bergmann et al., Eur. Phys. J. C (2019) 79: 165. <https://doi.org/10.1140/epjc/s10052-019-6673-z>



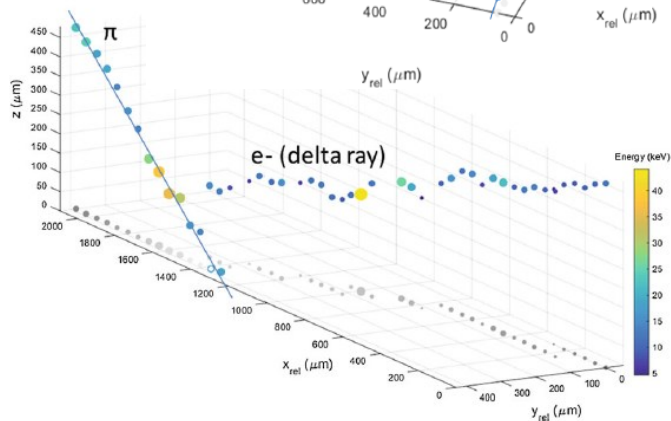
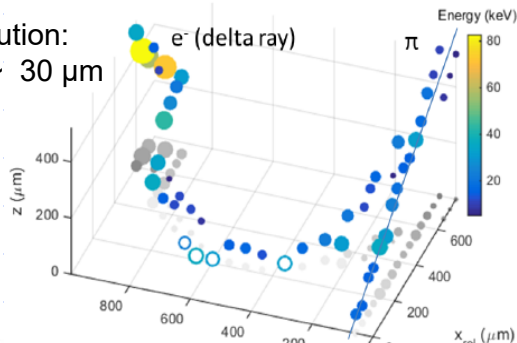
3D track reconstruction

Silicon sensor 500 μm

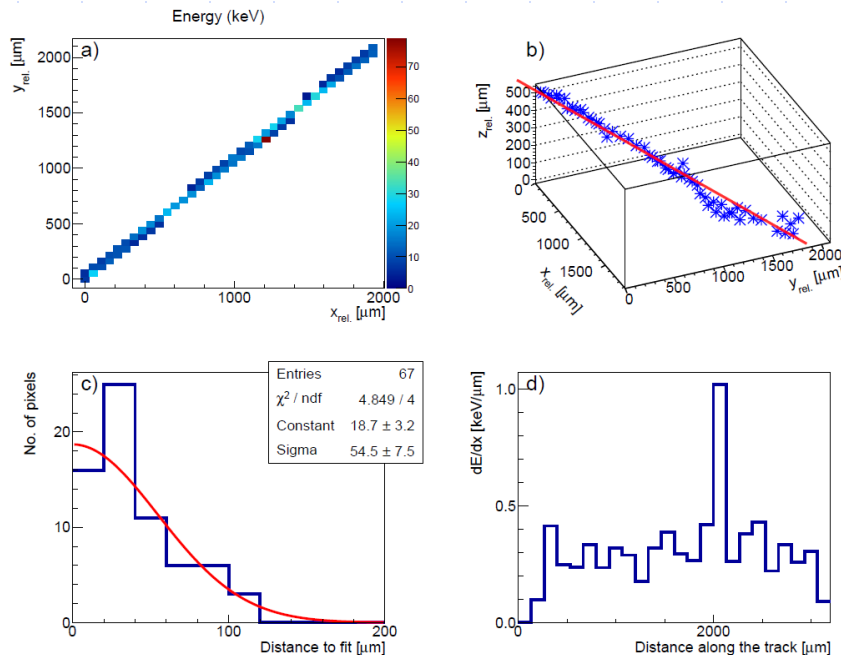
Test beam measurement

120 GeV/c pion tracks accompanied by δ -rays:

z-resolution:
 $\sigma_z \sim 30 \mu\text{m}$



A cosmic μ measured in the Prague laboratory:



Bergmann et al. Eur. Phys. J. C (2017) 77: 421.
<https://doi.org/10.1140/epjc/s10052-017-4993-4>



3D track reconstruction

CdTe sensor 2 mm

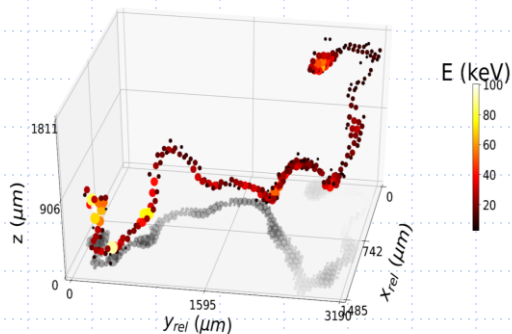
Test beam measurement 2



Trajectory reconstruction precision
<200 μm at 1 m distance

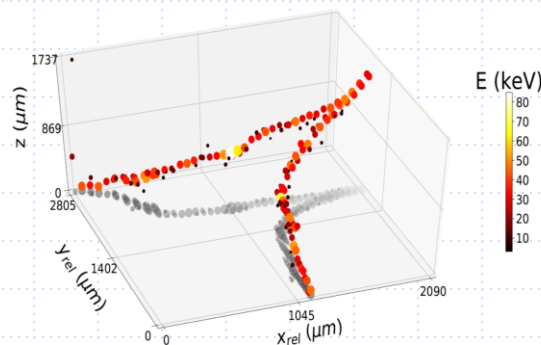
$E_{\text{dep}} = 5.66 \text{ MeV}$

e^-/μ^+ like event



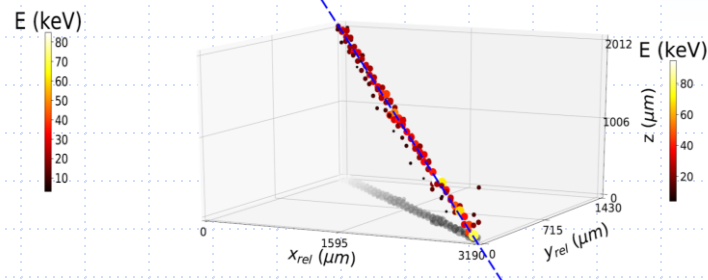
$E_{\text{dep}} = 3.96 \text{ MeV}$

pair production



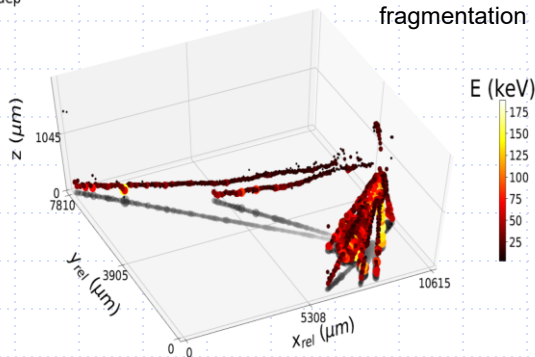
$\frac{dE}{dx} = 1.24 \frac{\text{MeVcm}^2}{\text{g}}$

cosmic μ



$E_{\text{dep}} = 37.06 \text{ MeV}$

fragmentation



z-resolution: $\sigma_z \sim 60 \mu\text{m}$

→ Improved determination of track directions

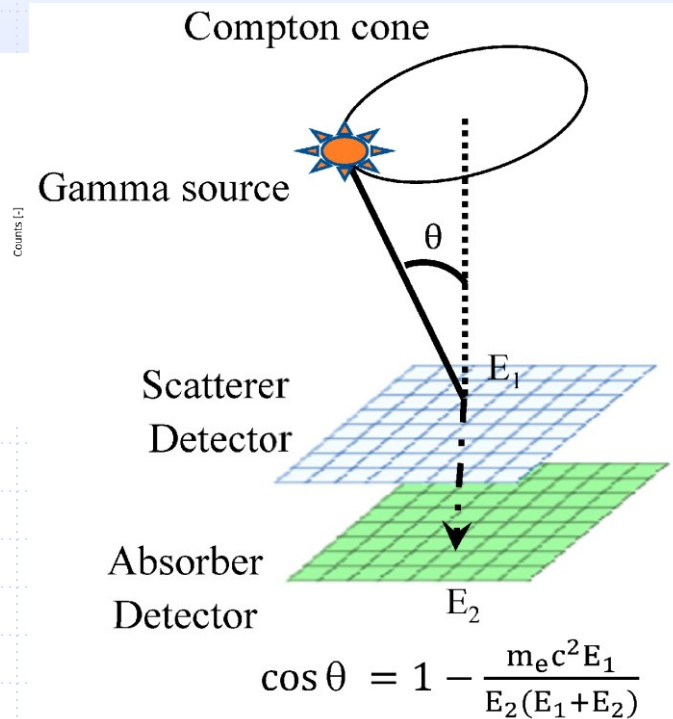
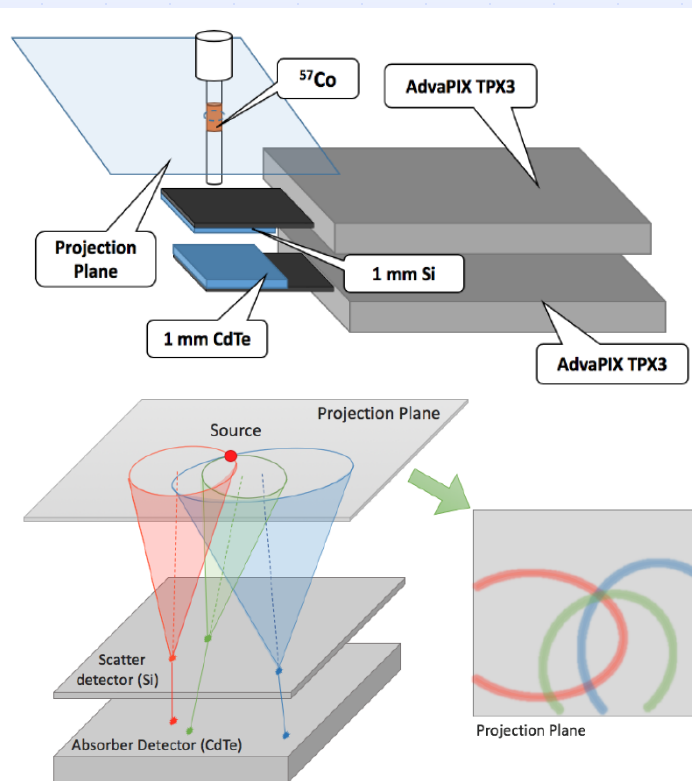
→ Improved separation of different particle classes

3D reconstruction enables the use as
single-layer Compton camera
(and polarimeter)



Timepix3 for coincidence gamma ray spectroscopy

Compton camera with CdTe and Si detectors

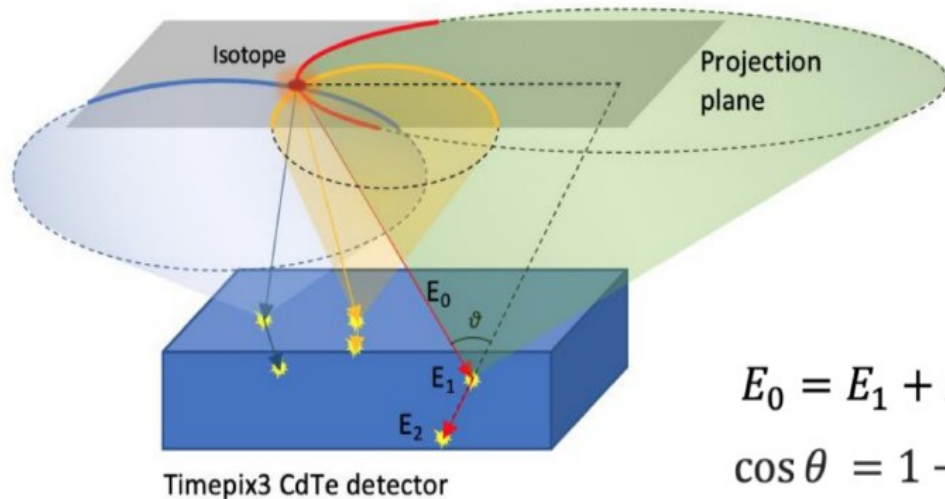




3D reconstruction

Application as a single-layer Compton camera

D. Turecek *et al* 2020 *JINST* 15 C01014



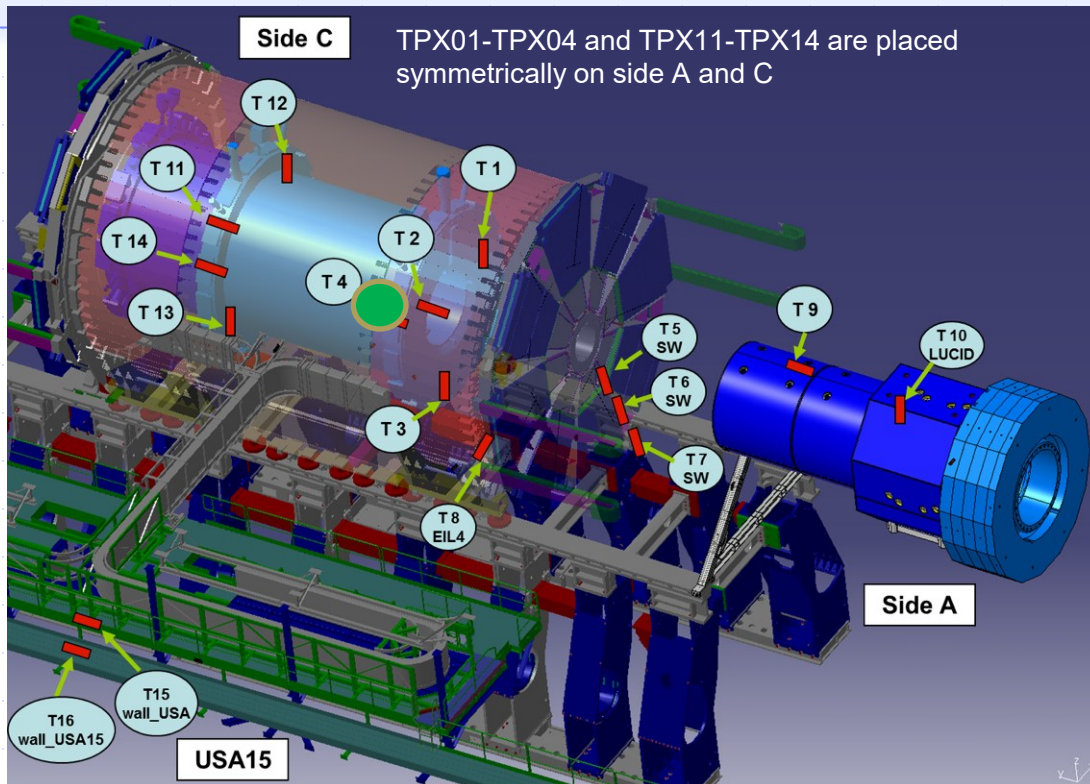
$$E_0 = E_1 + E_2$$
$$\cos \theta = 1 - m_e c^2 \frac{E_1}{E_2(E_1 + E_2)}$$

- Temporal coincidence detection of Compton electron and scattered photon
- Energy information for apex calculation through kinematics reconstruction
- Vector between interactions defines axis



ATLAS-TPX devices in the ATLAS experiment

Timepix3 in ATLAS is synchronized with LHC clock

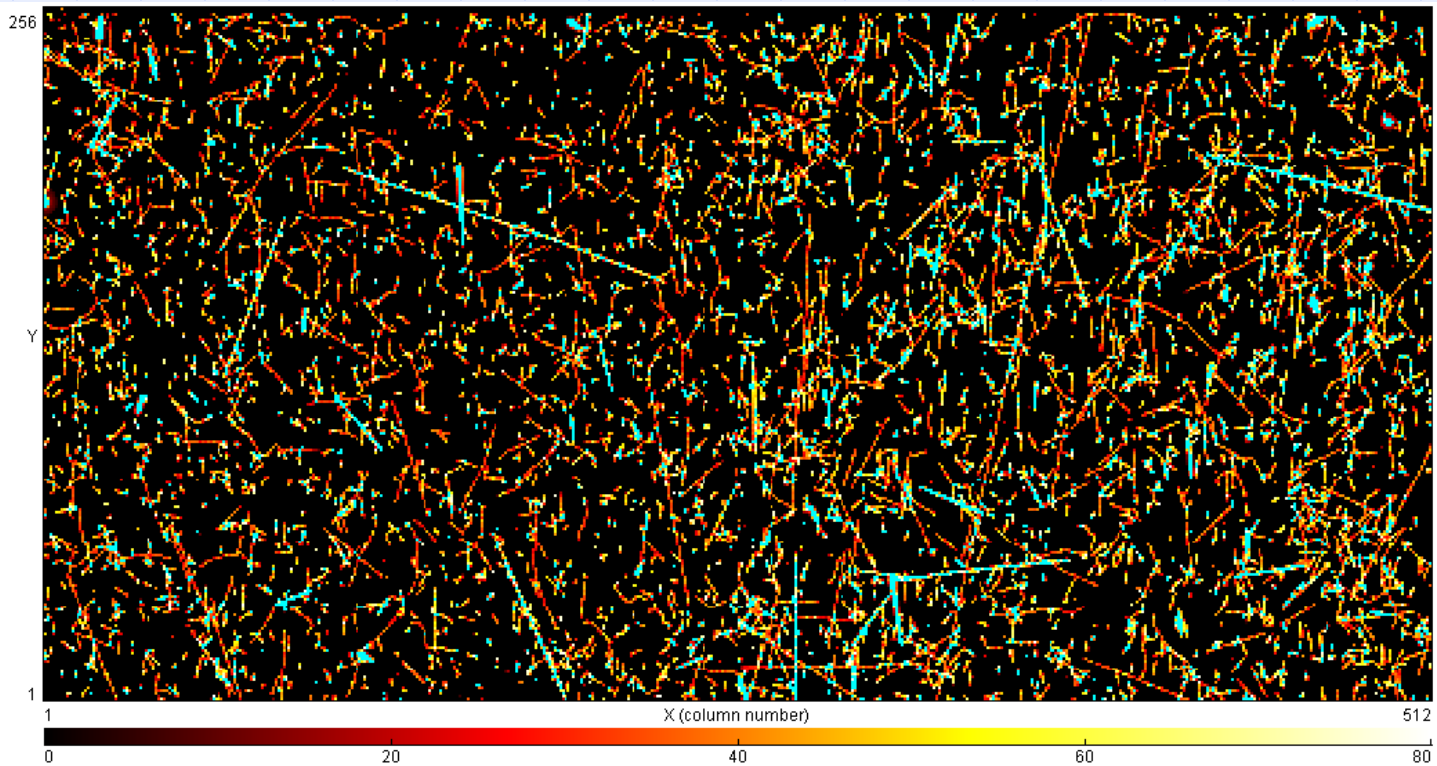


TPX-2013

TPX3-2018

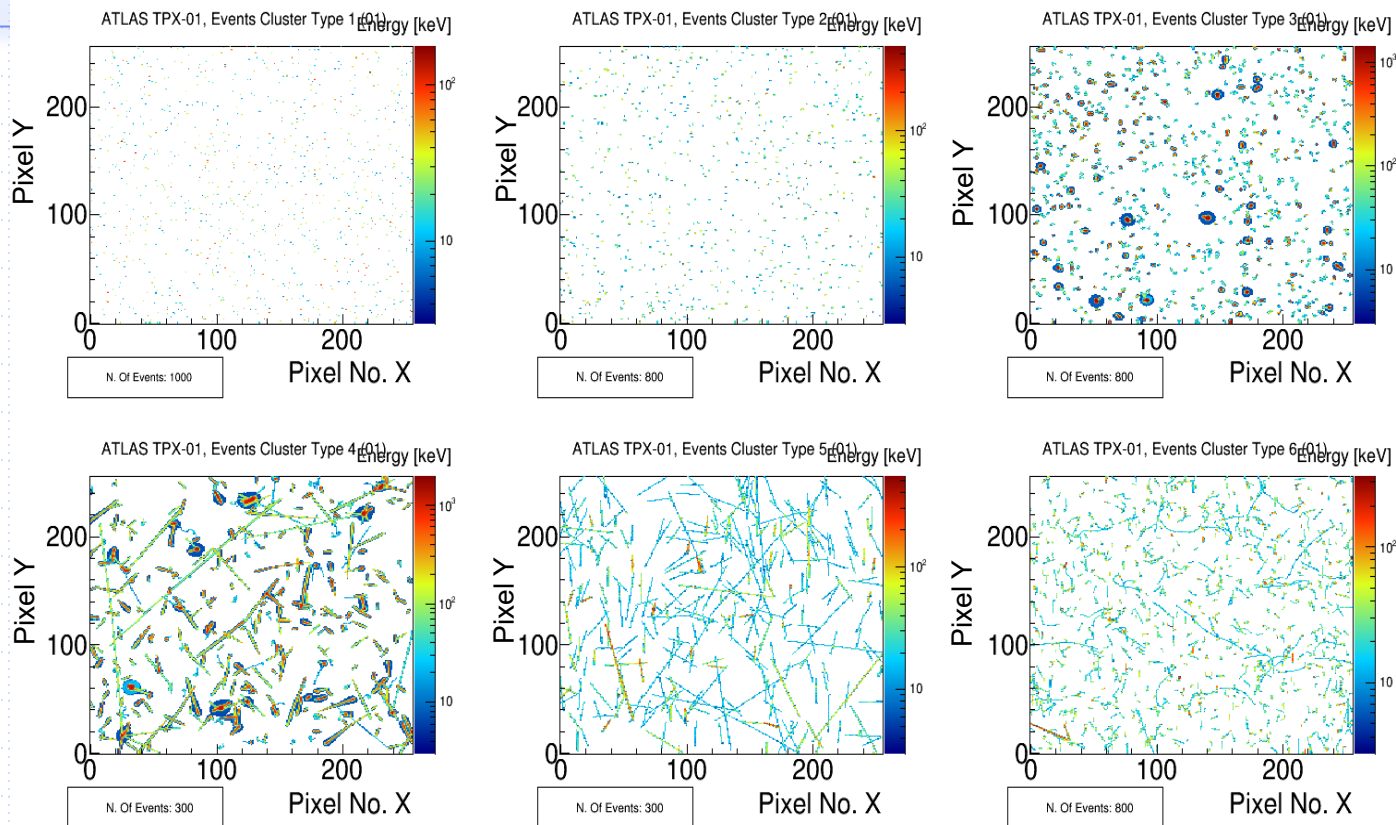


Response of the ATLAS-TPX device to isotropic mixed radiation field sandwich of 2 detectors, left 300 μ m, right 500 μ m





Sample of clusters measured during the collision period by ATLAS-TPX 01 (500 μm sensor) selected by means of developed pattern recognition procedure





Composition of radiation environment in Central barrel region of ATLAS experiment



60s „frame measured during the LHC beam-beam collisions



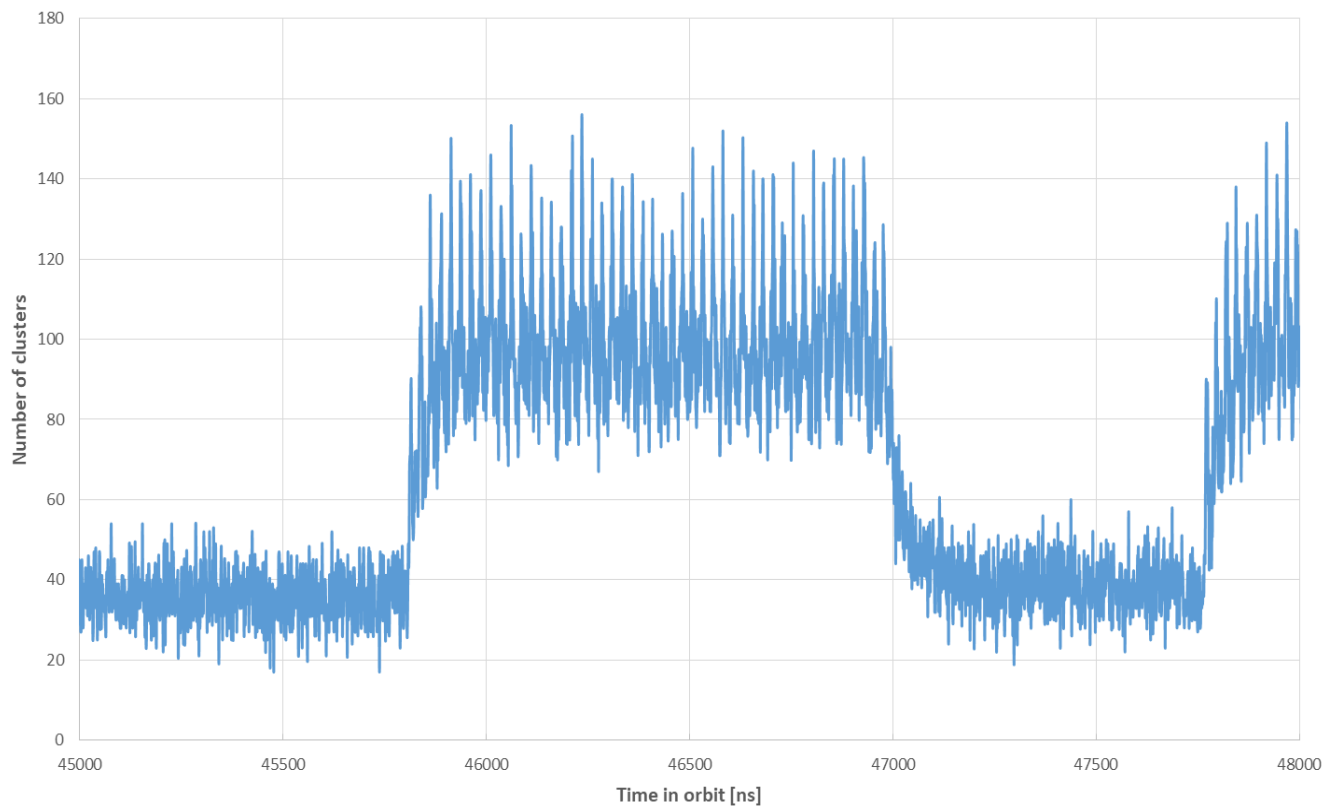
Time of arrival



Deposited energy

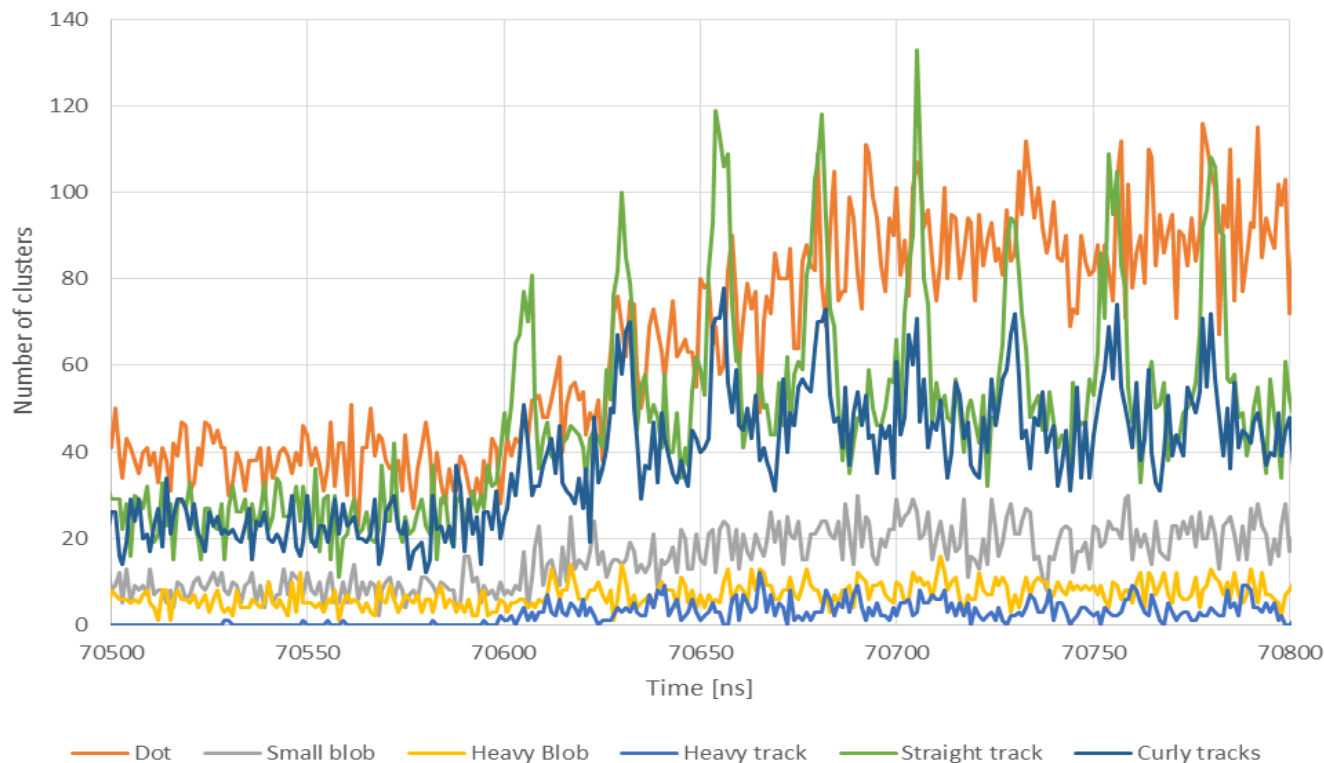


Detail view on bunch-bunch collisions recorded by ATLAS-TPX3 device synchronized with LHC



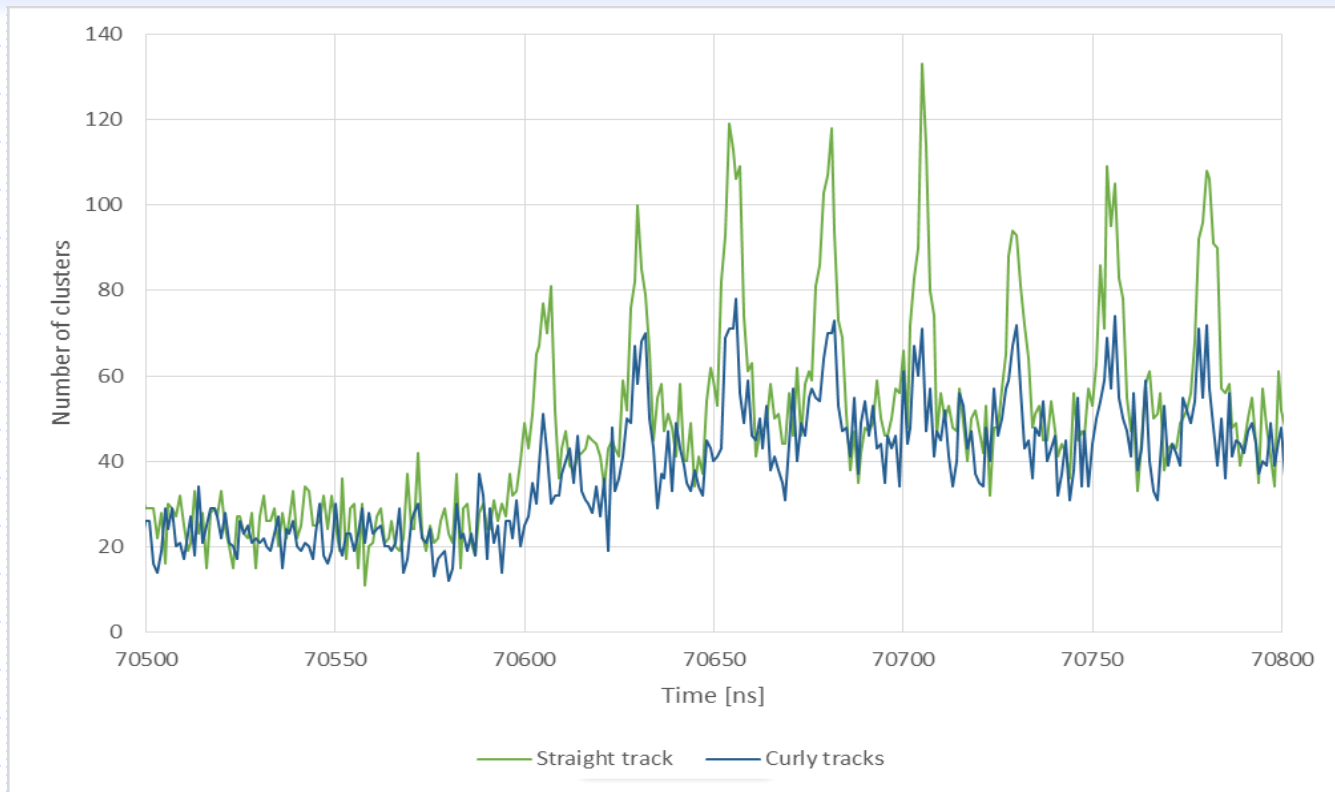


Composition of recorded clusters during individual bunch-bunch collisions including operational radiation background



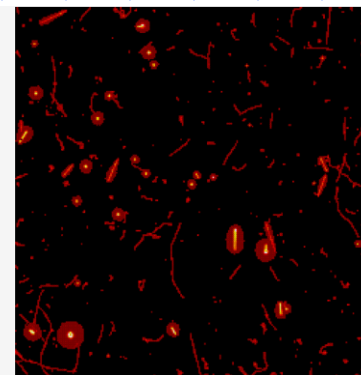
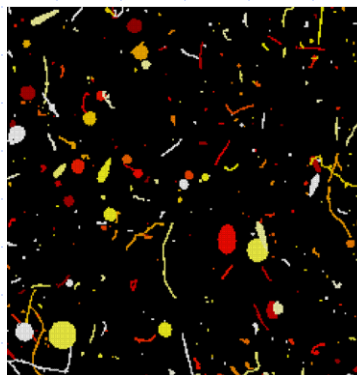
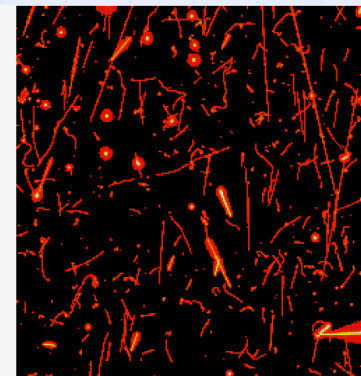
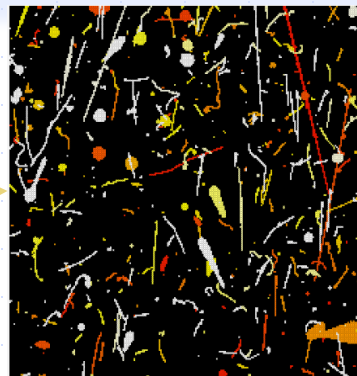
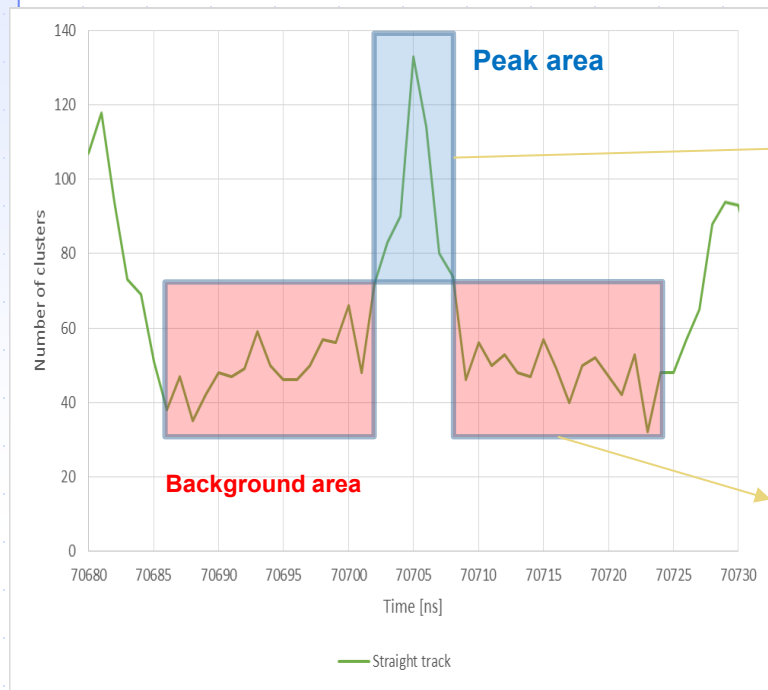


Straight tracks (MIPS) and curly tracks dominating periodically within 2.5 ns bunch-bunch collisions





Recorded tracks of particles generated during an individual bunch-bunch collision of the LHC beam superposed on radiation background in ATLAS

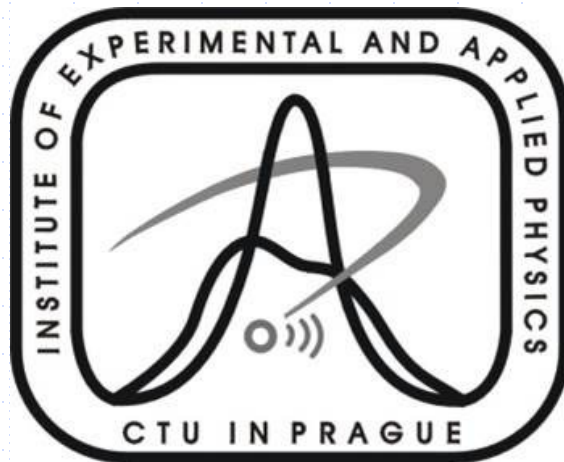


Time of arrival

Deposited energy



Thank you for your attention



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